

10/15/2025

Crossed Syndromes and Exercise Prescription



more integrative approach to treat

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Touro University California College of Osteopathic Medicine

PowerPoint and lecture built upon
the previous work of many faculty,
including most recently,

Stacey Pierce-Talsma D.O,
Jordan Keys D.O.

OBJECTIVES

1. Describe the **response of body tissues to gravitational strain pathophysiology (GPS)**
2. Identify the difference between a **structural lesion** and a **functional lesion**
3. Describe the components of the **Upper and Lower Crossed Syndrome**
4. **Identify muscles** involved in the upper & lower crossed syndrome (LCS) as **postural/tonic or phasic**
5. Delineate the key components **to treating muscle imbalance** as outlined by **Dr. Janda**
6. Outline the importance of addressing faulty movement patterns in the lower and upper quarter
7. Review muscle imbalance and its proposed causes
8. Apply understanding of the LCS and UCS to diagnostic findings and proper treatment given those findings
9. Identify the **6 most common dysfunctions in “failed lower back pain syndrome”**

Patient N.D.

- 24 yo male with CC of low back pain and outer hip pain. Started about 4 months ago- has progressively been getting worse. At least 1/10 at worst 5/10 sore and aching without radiation. No associated numbness, tingling or weakness. Worse with running, better with rest and stretching (which he rarely does). Has not tried any medications. Has never had this pain before. No history of injuries but he has recently been training for a marathon.
- PMHX: none Allergies: NKDA
- PSHX/Trauma: right ACL repair (after a skiing injury 2 years ago)
- Medications: none
- SHX: No ETOH, no Tobacco, single, sexually active with GF, no illicit
- Familyhx: no hx of inflammatory arthritis

What else do you want to know about our patient?

- Other HPI?
- What would you do for PE?
- What would you do for Treatment?



- VSS Normal, normal gait with over-pronation and right flattened arch, increased lumbar lordosis
- Abdominal Exam Normal
- Neuro: LE strength +5/5 in all Myotomes, Sensation intact to light touch in all Dermatomes, UE/LE DTR +2/4 no clonus, SLR negative
- Renal: no CVA Tenderness or hematuria
- MSK: normal LS ROM, no step off abnormalities, + increased lordosis, + anterior pelvic tilt, + hypertonic lumbar paraspinals, no scoliosis
- Osteopathic: Right innominate anterior, R on L sacral torsion, bilateral hamstring hypertonicity worse on the right, bilateral iliopsoas hypertonicity worse on the right, L5ERSI, L1-4NSIRr, right tibial torsion, right fibular head posterior, right cuboid dropped.

Patient N.D. returns to the clinic two weeks later...

- What would you do now?



- OMT was helpful and he had improved symptoms for several days
- Now aching and sore again

Let's think about our patient and review some postural considerations as well as some new ideas...

look deeper and see



Postural Relationship to Gravity

sometime send pt to pool bc gravity not pulling anymore

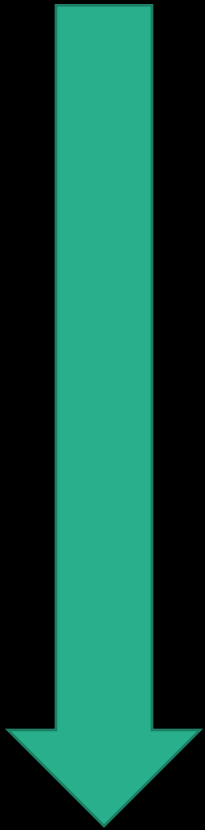
studying - get up and stretch - can compensate
if injury - cannot compensate = strain

Gravity- constant stressor

Posture- distribution of body mass in
relation to gravity over a base of support

Muscle balance continually adapts to the
body's posture- mechanical responses and
adaptations **Strain** is a maladaptation

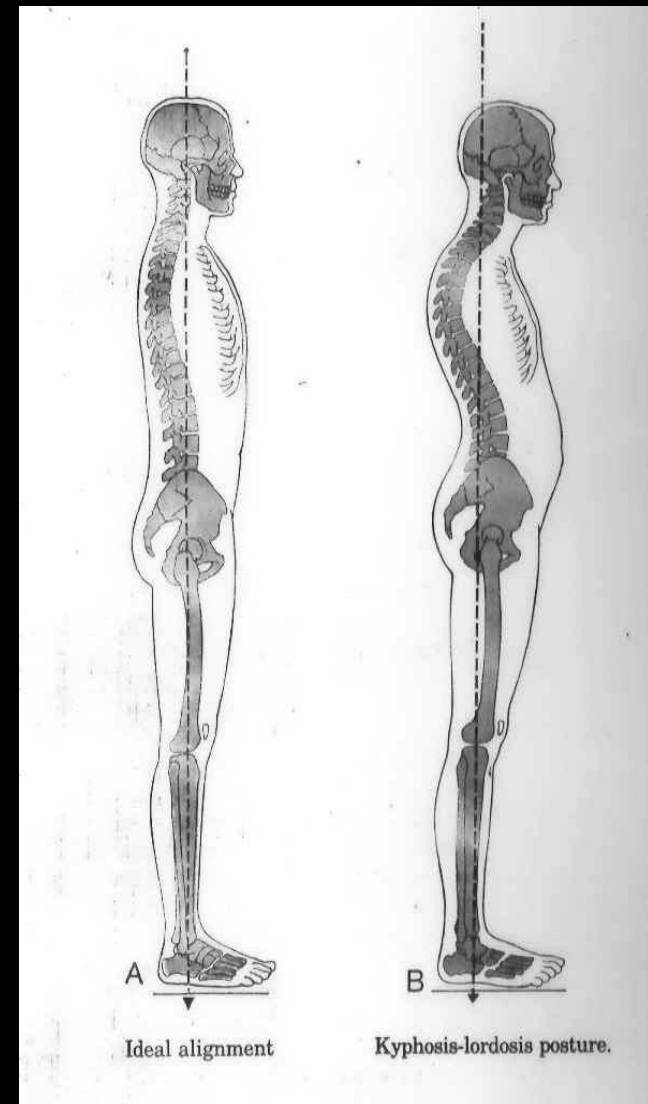
Chronic postural functional
demands will accelerate
breakdown/Decompensation



Gravitational Strain Pathophysiology (GPS) and Postural Decompensation

- **Posture**
 - Distribution of body mass in relation to gravity over a base of support
 - Any compensation will require counterbalancing
 - Muscles must adapt
- **Gravitational Strain Pathophysiology**
 - Strain from the body's attempts to stave off the effects of gravity
 - Leads to altered postural alignment and recurrent somatic dysfunction
 - Intrinsic and Extrinsic factors
- **Postural Decompensation**
 - Homeostatic mechanisms are overwhelmed
 - Pathological change becomes structurally incapable of resisting gravitational forces

gravity only goes one way so if head goes forward will continue to do so and will cause postural compensation



Gravitational Strain Pathophysiology (GPS)

- **INTRINSIC**

- Age (muscle weakness, fascial elasticity, bone mass)
- Altered integrity of soft tissue (hypermobility, CT disorder) influence how system can manage forces
- Incompetent bony structures (spondylolysis, spondylolisthesis)
- Poor Tissue Health (disease, poor nutrition, smoking)

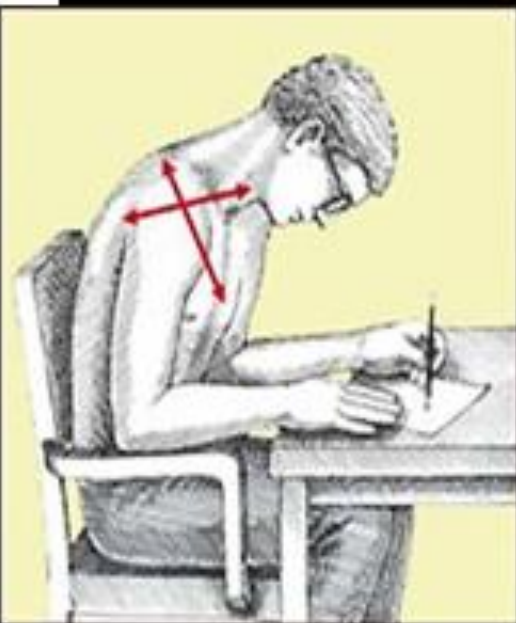
change kyphotic/lordotic curve

- Altered base of support (flat feet, high heels, short leg, altered gait after injury or surgery)
- Change in body habitus (pregnancy, weight gain)
- Ergonomic factors (how to people sit/stand/walk)

over/under training

- **EXTRINSIC**

- Surgeries
- Traumas
- Increased Gravitational stress (like a G6)



pregnancy



kyphosis - adding ergonomic will change :

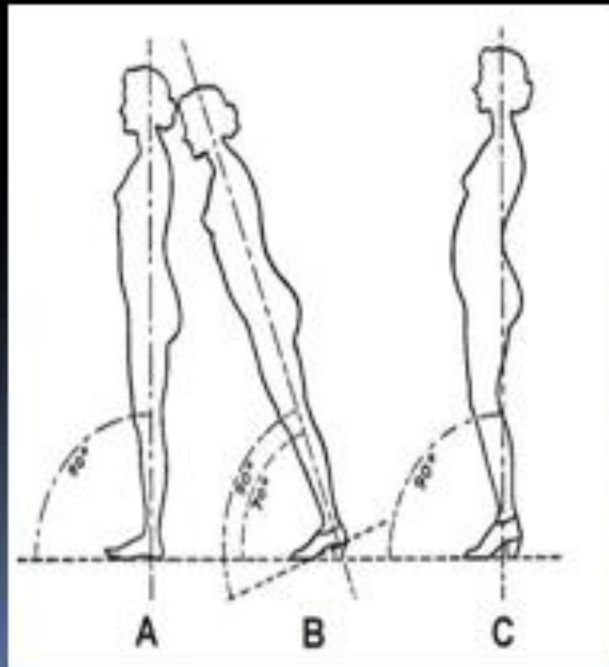
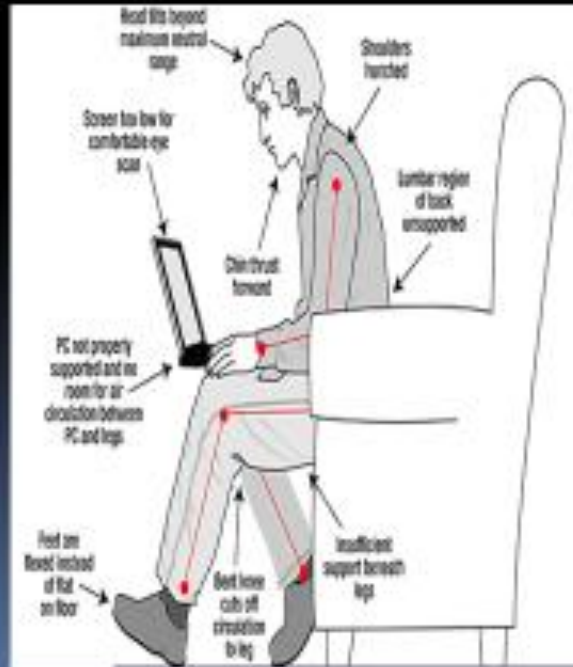


Fig. 3 © 2012 www.enkdallan.com

Posture and Mind, Body, Spirit

“Shapes are the way in which we know who we are and where we are in our universe. Show me the shapes and forms a man gives to his life, and I will tell you whether he is a master or victim of that life.”

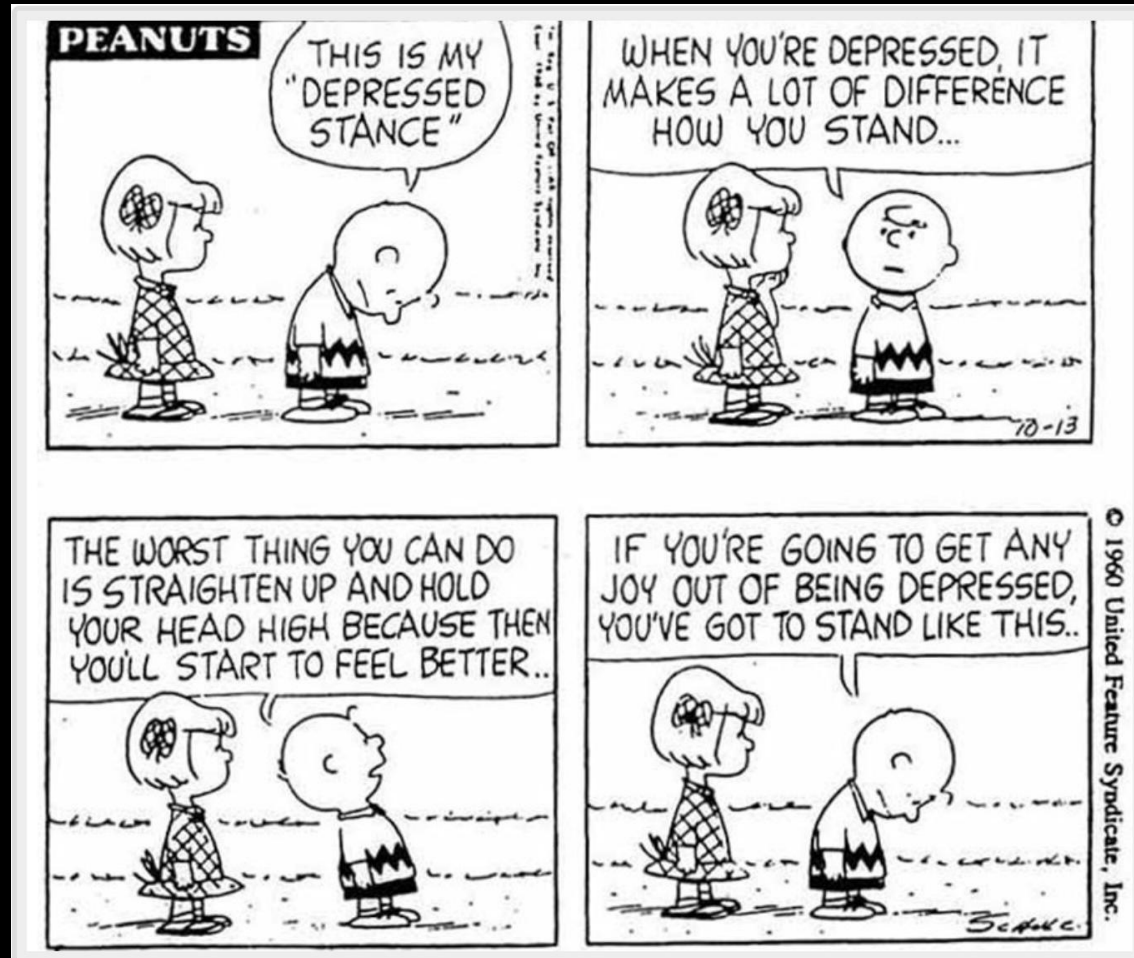
• Gail Godwin, Glass People

- Emotions take shape through anatomy and all of our anatomy is hard wired into our neurocortex

• Sensorimotor system

- Our emotions change our shape
- Can our shape change our emotions?

posture reflect mind, body spirit?
can shape reflect emotion and vice versa?



Embodied Cognition

- **ATTITUDE- Miriam Webster**

- “The Arrangement of the parts of a body or figure: Posture”

- **Slumped Vs. Upright Posture**

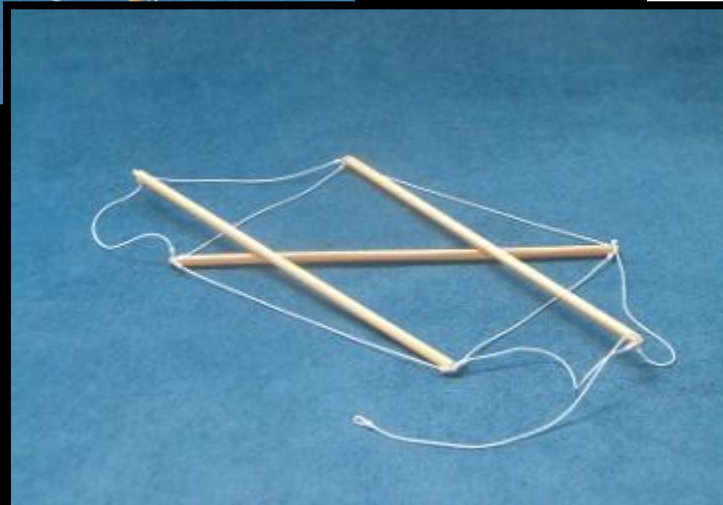
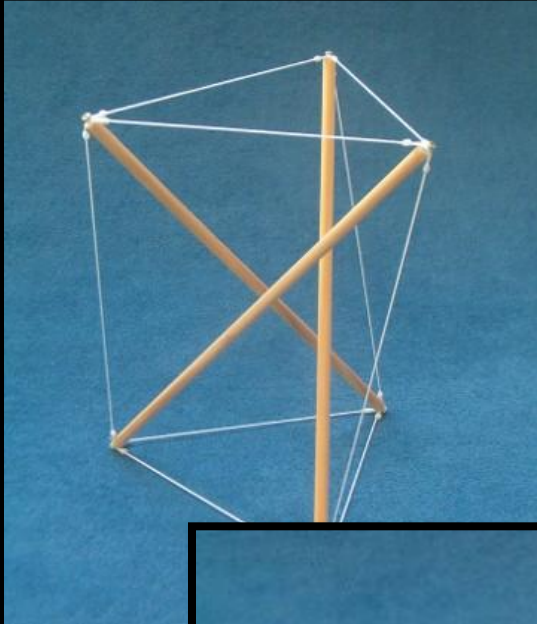
- Effect on Confidence, metacognition
- Posture can affect direction of thoughts
- Slumped- lower self esteem and mood and greater fear
- Influences Productivity
- Influences Memory
- Upward head tilting & Posture increased pride
- Poor posture can leave you vulnerable to crime
- Slumped posture led to helplessness & lack of persistence
- Movement in a “yes” pattern produced increased preferences

‘Sometimes your joy is the source of your smile, but sometimes your smile can be the source of your joy”

- Thich Nhat Hanh

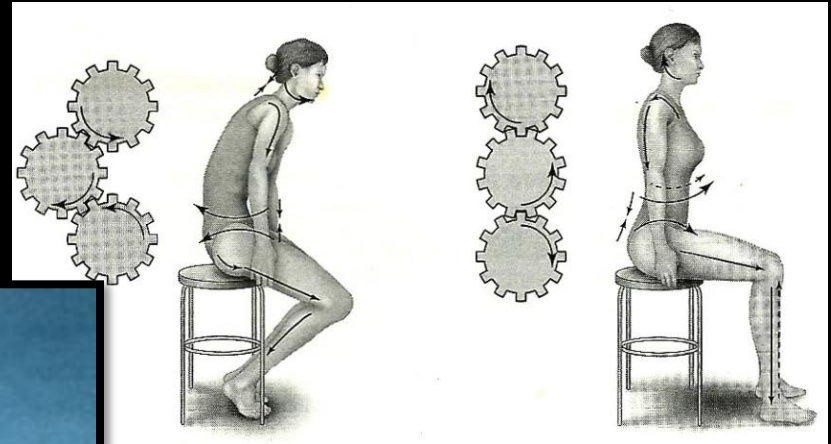
when M smiles the treatment changes even if pt can't see

Tensegrity & Posture



allostatically more challenging
becomes a functional place of rest

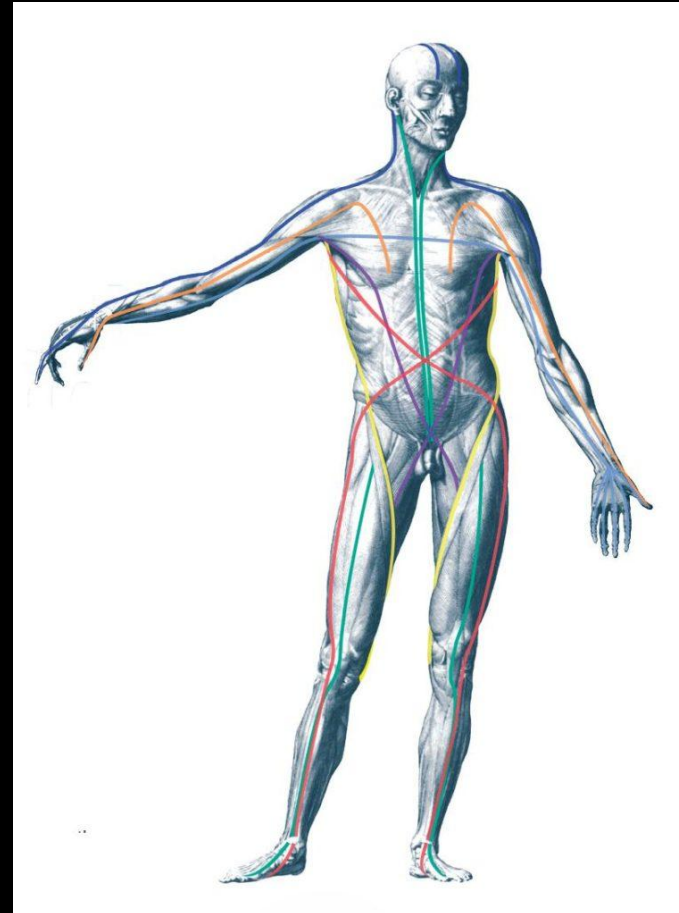
makes work sitting so much
less



head forward has no choice but to cause kyphosis

Interconnection

- Tensegrity
- Anatomy can no longer be origins and insertions
- Functional anatomy
 - Interconnected muscle groups
 - Tensioned fascia
 - Muscle balance
 - Joint protection
 - Neural patterning



Structural Lesions VS Functional Lesions

Structural Lesions-

- rooted in anatomy and biomechanics- structural approach to pathology
- relying in imaging- damages to physical structures- this approach is foundational to medical education and practice
- Diagnostic tests may be inconclusive- more than likely there is a functional cause.

Functional Lesions –

- impairment in the ability of a structure or system to perform it's task
- impairment manifests as reflexive changes
- much more difficult to diagnose and treat- you have to understand complex interactions within the system
- Shifts our focus from structure to function

reflexive changes and they become chronic
x ray may not change but is good to keep track of

**FOCUS ON THE CAUSE OF THE PROBLEM RATHER THAN FOCUS
ON THE PATHOLOGY**



<https://www.linkedin.com/pulse/20140917125222-154514887-cervical-spondylosis-knowing-what-it-is-a-way-out>

Structural Lesions on Imaging

- Disc herniations are seen in 22-67% of asymptomatic adults and spinal stenosis in 21%, of **asymptomatic adults** over age 60
- One systematic review showed 90% over 60 had a disc herniation or bulge that was asymptomatic
- Osteoarthritis often seen **but poorly correlates** with symptoms
- Study of 188 people age 40-80- 60% males & 67% females had facet joint OA on lumbar CT- **no correlation with back pain**
- Even if the structure is causing the problem- **unclear if it correlates to the clinical severity and outcome**

imaging not appropriate on it's own to make a dx

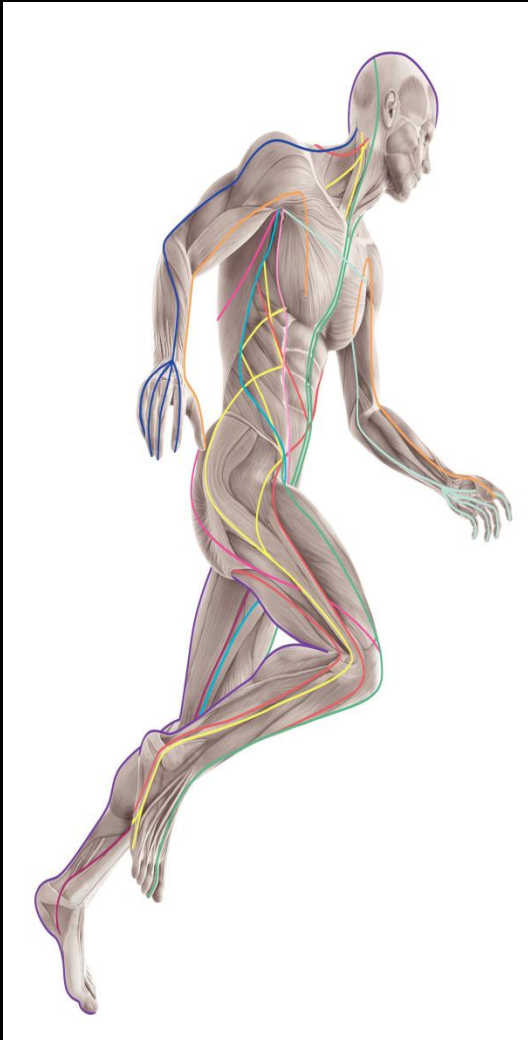
“In patients with chronic musculoskeletal pain, the **source** of the pain is **rarely** the actual **cause** of the pain.”

guy on L doing all the work but guy on R is saying



Assessment and Treatment of Muscle Imbalance- the Janda Approach

Vertical & Horizontal Adaptation



- No biomechanical lesion remains localized. Its effect can be on other muscles and joint dysfunctions above or below the original site.
- **Vertical Adaptation**- motor re-programming- muscle firing patterns
- **Horizontal adaptation**- one joint problem leads to joint problems in the next joint up ACL injury and horizontal adaptation causes issue above
- The more distal the injury the greater the proximal trunk disorganization.

Gravitational Strain and Postural Decomensation affects:

Bones, Muscles,
Ligaments, Fascia,
Vasculature, Connective
Tissue, and Organs

Response of Tissues to Gravitational Strain and Postural Decompensation

• Skeletal-Arthroidial Response

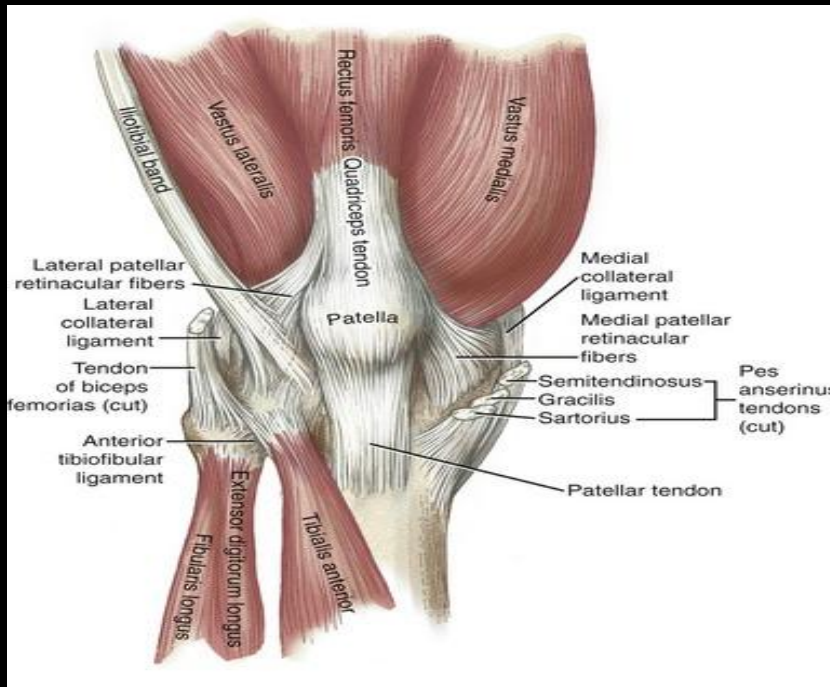
- Wolffe's Law
 - Bone remodels according to stress
- Spurs, wedging, and degenerative changes

• Ligamentous Response

- Stabilizing ligaments become stressed
 - Iliolumbar
 - Sacrotuberous
 - Long dorsal sacroiliac ligaments

- Can become Tender
- Calcified remodeling
- Sclerotomal pain pattern

pain from attachment of ligament or tendon to bones



Response of Tissues to Gravitational Strain and Postural Decomensation

■ Muscular Response

- Elastic component
 - Transient functional change in connective tissue length
- Plastic component
 - More permanent structural deformation in response to static postural change
 - Shortening not responsive to Muscle spindle resetting (ie. Counterstrain)
 - Fibrosis hardening and thickening of tissues
 - Fatty Replacement
 - Myofascial remodeling along vectors of postural stress

- Trigger points & tendonitis develop in muscles that tend to be weak and inhibited.

muscle can't support force coming through

- Tears and ruptures develop in muscles that tend to tightness.

- Avulsion fractures and apophysitis common at attachments of muscles that tend to tightness.

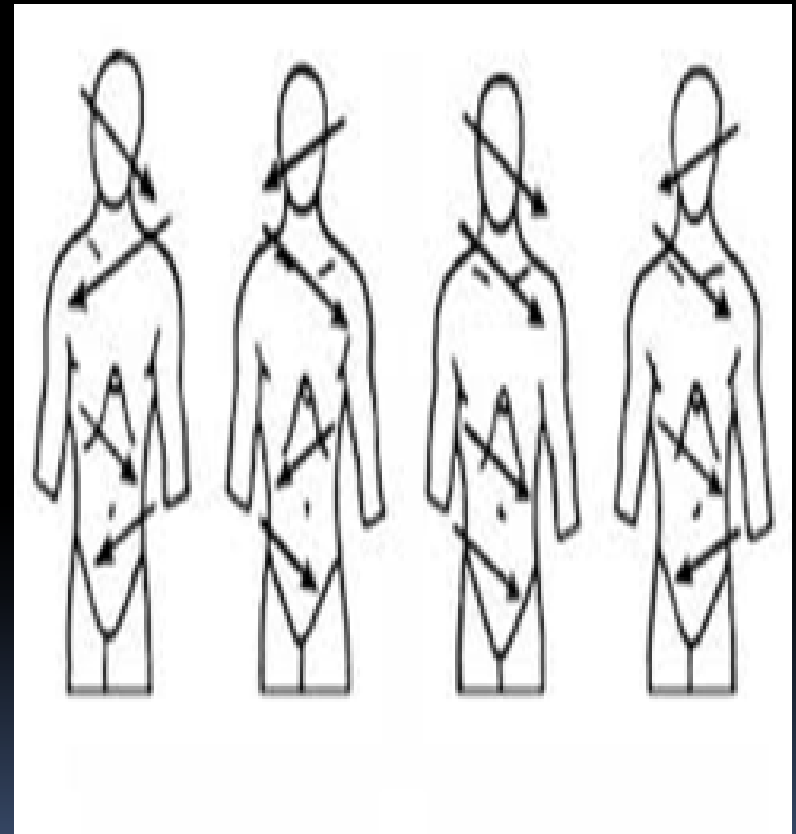
force through tendon onto bone

- Sacroiliitis and Osteitis Pubis not likely 1° inflammatory but imbalance from wear and tear

forces arriving at pelvis

Response of Tissues to Gravitational Strain and Postural Decomensation

- Neural, Vascular, Lymphatic and Visceral Element Response
 - Transition zones in the body related to transverse diaphragms
 - Dysfunction leads to decreased lymphatic and venous return
 - Posture can relate to **autonomics** and facilitated segments affecting organs

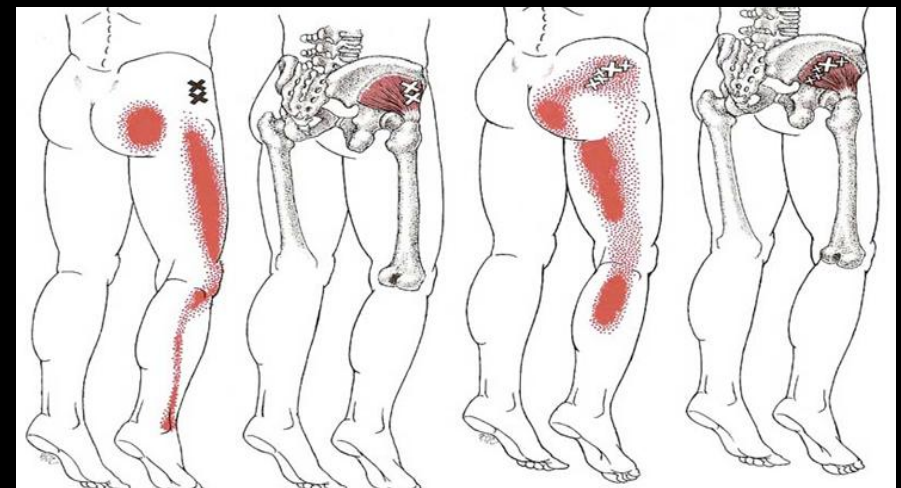
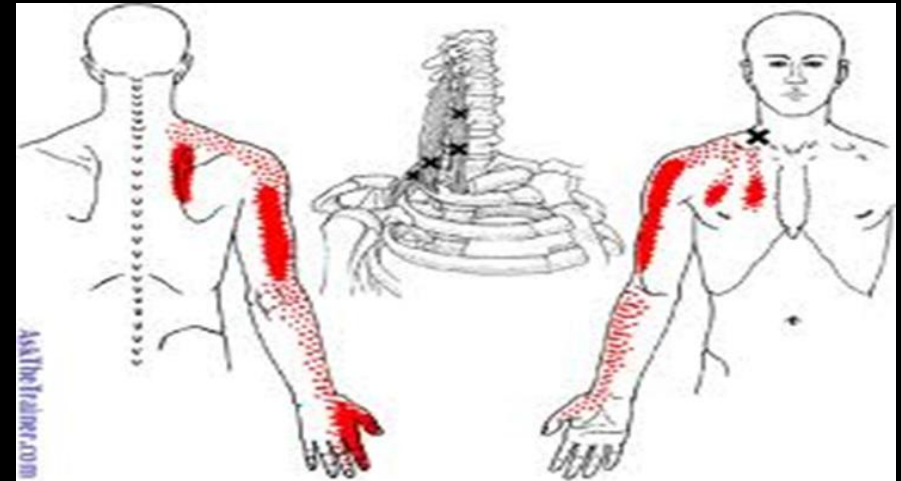
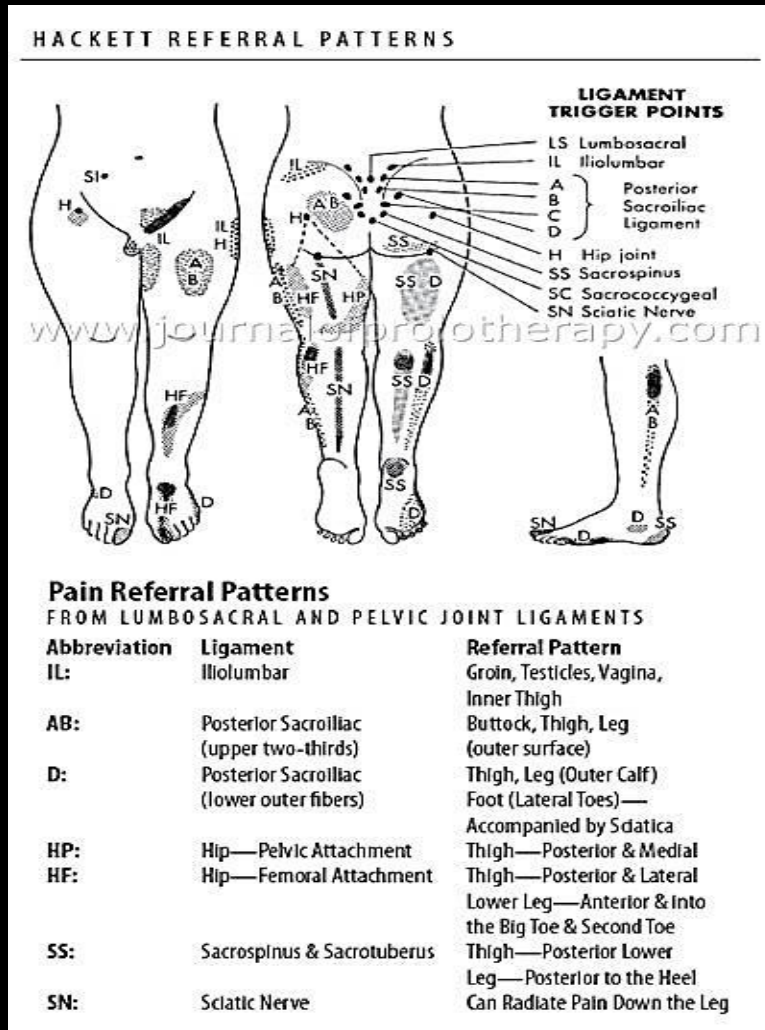


just bc of posture

Zink is myofascial screen www.erikdalton.com

If you don't treat the cause, you may end up chasing the symptoms...

dysuria - not UTI but is pelvic floor instability



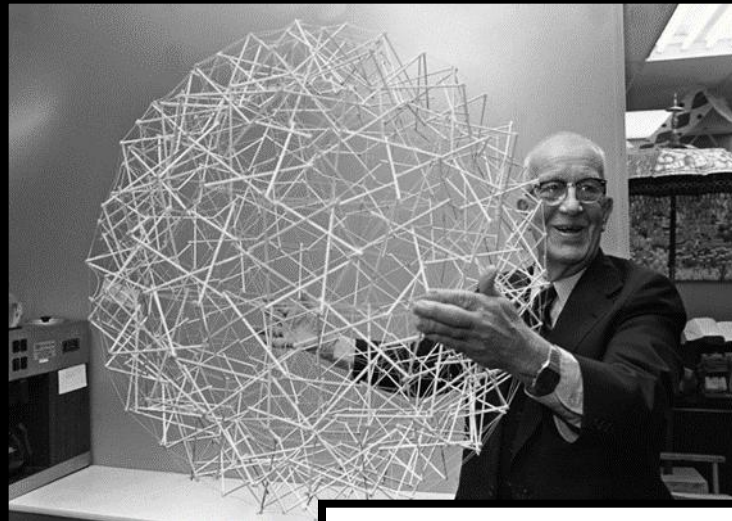
from muscle itself

Sclerotomal Referral Patterns

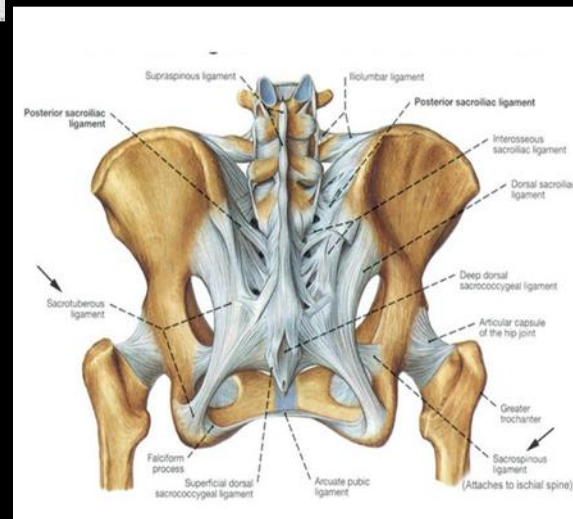
Trigger Point Referral Patterns

So How Can We Complete Assessment and Evaluation Looking for GPS and Postural Decompensation?

pain in SI - is SI itself or from anatomy trains?



“Even minor changes in one body region may affect significant biomechanical, tensile and ergonomic changes elsewhere”



Anatomy Trains by Thomas Myers

Vladimir Janda, DMed, DSc



Dr. Greenman & Dr. Janda

- 1928-2002
- Chair Department of Rehabilitation Medicine, Charles University, Prague, Czech Republic
- **Muscle Response to impaired afference is not a random occurrence**
 - Sensorimotor System
 - Sensory changes lead to an altered motor program
 - Responds in a predictable way based on structure and function
 - Biomechanical changes vs neurological changes
 - Importance of proprioception

muscle is crossroads of adaptation

Assessment and Treatment of Muscle Imbalance

*The Janda
Approach*

Phil Page
Clare C. Frank
Robert Lardner

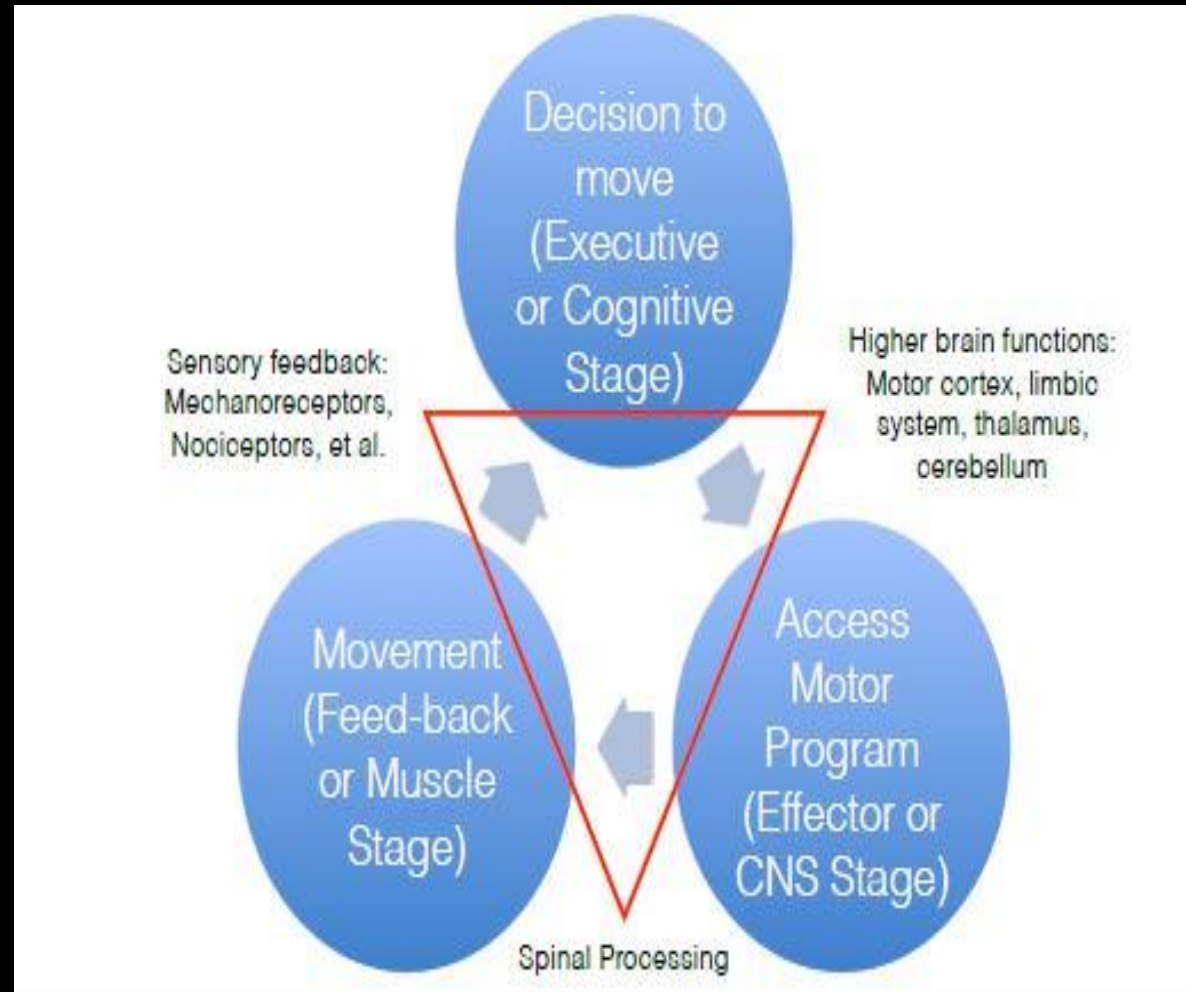


ppl who trained with Dr. Janda

are patterns you can recognize

How do we move...

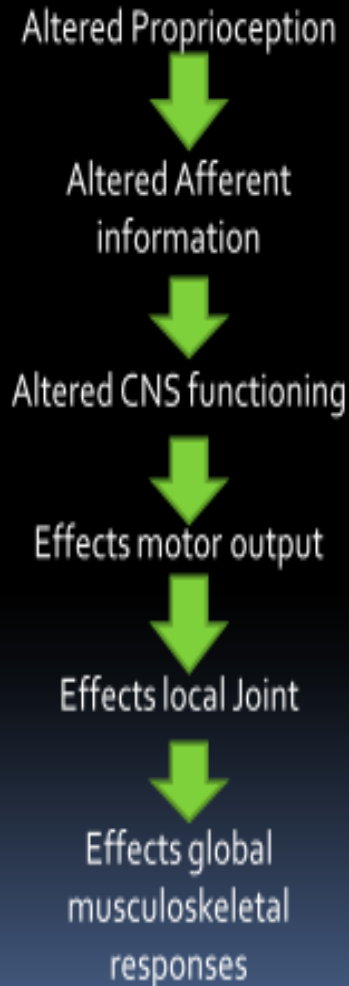
Somatic dysfunction is a form of abnormal motor control for which the body must compensate. It is neurologically mediated and a manifestation of the entire central nervous system.



just treating one is not enough need to treat the whole system

Slide from Dr. Richard Schuster DO

The Sensorimotor System



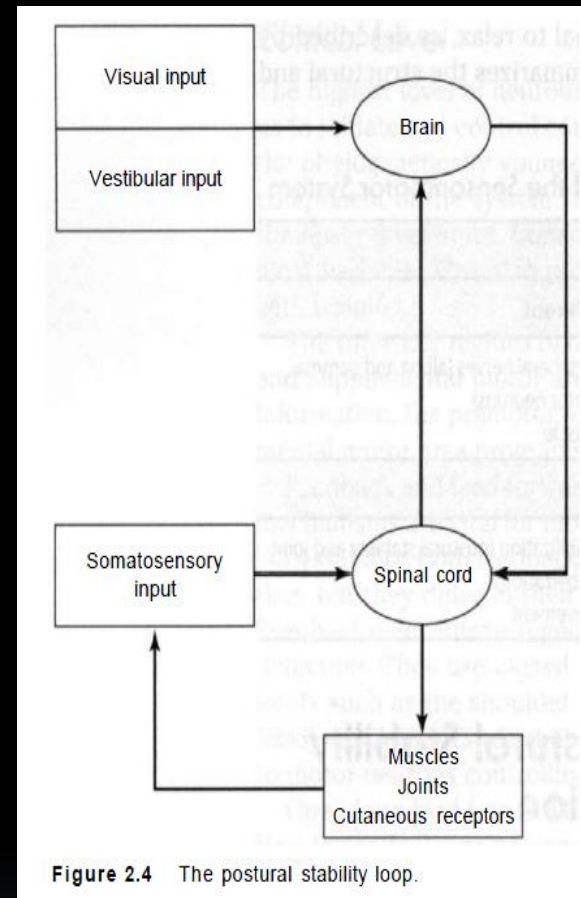
- "The sensory system is the key to proper function of the motor system"
- "clinicians must consider the whole body in sensorimotor dysfunction"

3 Key Areas of proprioception for posture

1. Foot 2. SI Joint 3. Cervical Spine

Pain is the strongest stimuli of this system

Poor proprioception- we recruit muscles harder

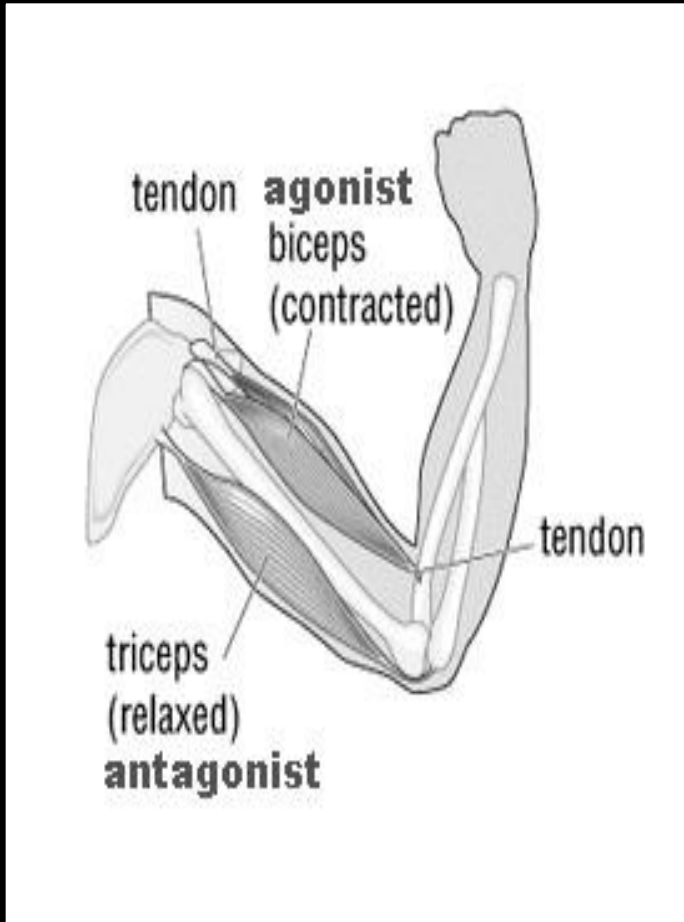


[http://blog.xuite.net/iamnig/vhplreh/301770399-Postural+Stabilization.](http://blog.xuite.net/iamnig/vhplreh/301770399-Postural+Stabilization)

Muscle Imbalance Description

- **Alteration in Length, Strength and Motor Control**
- Balance needed for opposing muscle groups to be coordinated
- Length or strength changes of agonist and antagonist muscle prevents normal function
 - **Muscle Length**
 - Facilitation/tightness/shortening of postural/tonic muscles
 - **Muscle Strength**
 - Inhibition/pseudoparesis/weakness of phasic muscles
 - **Motor Control**
 - Loss of integrated coordinated neurologic function
 - Incorrect sequence of muscle firing patterns

Agonist/ Antagonist Reflex



- Co-Contraction
 - Stimulation to both flexor and extensor muscles to stabilize a joint
- Joint Flexion
 - Flexor contracts, then via reciprocal inhibition extensor muscle group relaxes
- Joint Extension
 - Extensor group contracts, then reciprocal inhibition flexor muscle group relaxes
- The harder a muscle contracts the more it inhibits its antagonist
- Short/Facilitated muscles can inhibit their antagonist
 - Pseudoparesis

bicep = agonist, tricep = antagonist contract bicep but tricep needs need flexors and extensors to stabilize extensor cross pattern reflex as well

Agonist/Antagonist Reflexes

KNOW

Most Common Pairings:

1. Biceps – Triceps
2. Deltoids – Latissimus Dorsi
3. Pectoralis Major – Trapezius/Rhomboids
4. Iliopsoas – Gluteus Maximus
5. Quadriceps – Hamstrings
6. Hip Adductor – Gluteus Medius
7. Tibialis Anterior – Gastrocnemius
8. Anterior Deltoid – Levator Scapula
9. Forearm Flexors – Forearm Extensors

How Does Muscle Imbalance happen?

- **Disturbance of muscular function**- mechanoreceptors/nociceptors stimulated (altered posture and attempt to return to homeostasis)



- **Afferent** neural activity goes to the cord, brainstem and cortex



- Final common **efferent** pathway is the **alpha motor neuron** which causes the muscle fibers to contract (descending cortical signals have to be modified from sensory input)



- Then through the gamma system **muscle spindles adapt** leading to alteration in muscle tone



- Perpetuation of this loop leads to **ongoing aberrant muscle tone**

chronic increased tone tricep with chronic decreased tone of bicep

Muscle Imbalance- Neuroreflexive Component

- “**Faulty posture** results in alteration of the center of gravity, which initiates **mechanical responses** requiring muscle adaptation.”
- “**Change in a joint’s mechanical behavior causes neuroreflexive** alteration in muscle function through aberrant afferent mechanoreceptor stimulation of articular reflexes. “
- “Long term activation of **abnormal articular reflexes** **causes change in cord memory** from a normal to an abnormal adaptive program resulting in muscle imbalance.”

–Greenman, *Principles of Manual Medicine 3rd edition*

Altered
Proprioception



Altered Joint
Position



Joint position
becomes
embedded in CNS
as a motor pattern



Chronic changes

Theoretical Causes of Muscle Imbalances

1. Postural adaptation to gravity

1. Faulty posture shifts center of gravity and body tries to compensate

lifestyle

2. Neuro-reflexive due to joint blockage

1. Dysfunction of normal articular reflexes

3. Central nervous system mal-regulation

1. Limbic system
2. Loss of reciprocal Agonist/antagonist interplay

4. Response to painful or noxious stimuli.

1. Spinal cord level- facilitation
2. Higher centers- central sensitization

5. Response to physical demands

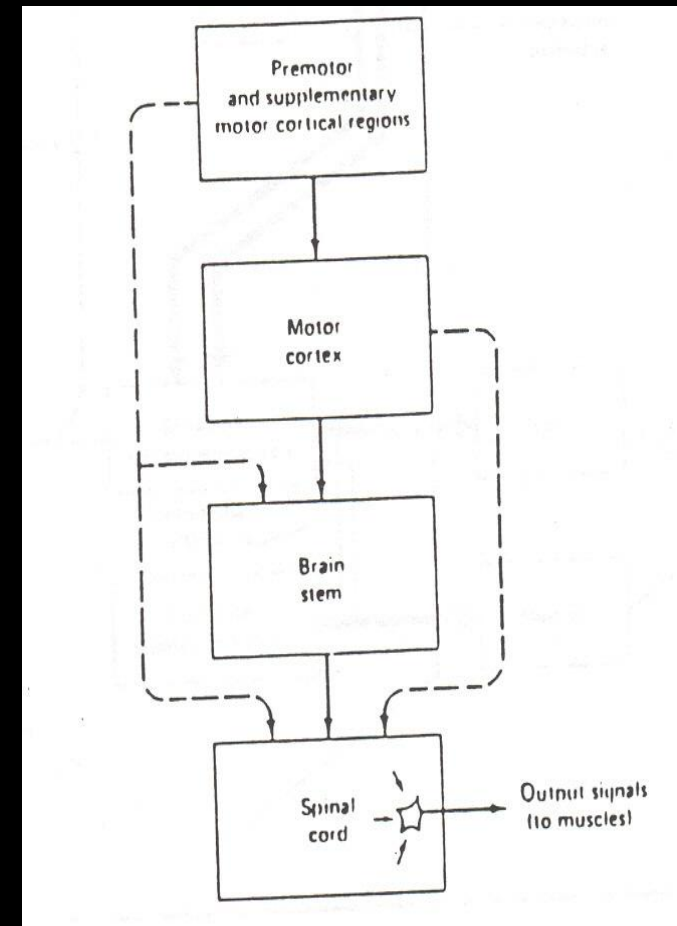
1. Fatigue

6. Habitual movement patterns

1. Repetitive stress, insufficient movement,
2. Repetitive movement

7. Psychological influences

1. Mood, somatization of stress, fatigue, pain



Lifestyle & Muscle Imbalance

- Lifestyle can contribute to muscle imbalance
 - Stress
 - Fatigue
 - Insufficient movement
 - Lack of variety of movement
 - Most repetitive movements activate the postural system neglecting the phasic system leading to imbalance
 - Bracing and anticipation of pain
 - Yoga, Pilates, and exercise retraining connection
 - anything that requires new movement

Muscle Imbalance

- **Begin by observing posture**
 - Occur in two major body regions
 - Pelvic girdle (lower crossed)
 - Shoulder girdle (upper crossed)
 - Alteration in firing pattern in principle muscle movements
 - Hip extension
 - Hip abduction
 - Shoulder abduction
- **Dynamic/Phasic Muscles**
 - Smaller Muscle Groups
 - Respond by
 - Inhibition
 - Hypotonicity
 - Weakness (Pseudoparesis)
 - **Postural/Tonic Muscles**
 - Large Muscle Groups
 - Respond by
 - Facilitation
 - Hypertonicity
 - Shortening

Tonic/Postural VS Phasic in the Lower Extremity

TONIC/POSTURAL

- Slow Twitch
 - Oxidative
 - More capillaries-Red fibers
 - Phylogenetically older
 - Typically Flexor or postural muscles
 - Typically one joint muscles
- Hip/Pelvis
 - Hamstrings, Iliopsoas, rectus femoris, TFL, Thigh Adductors, piriformis,
- Lspine
 - Erector spinae group (thoraco-lumbar extensors)
 - QL
- Lower extremity
 - Gastrocnemius and soleus
- SHORT, TIGHT ,FACILITATED with repetitive stress

PHASIC

- Fast Twitch
 - Glycolytic pathway
 - Low capillaries-White fibers
 - Phylogenetically younger
 - Typically extensor muscles
 - Typically two-joint muscles
- Hip/ Pelvis
 - Vastus medialis, intermedialis, lateralis, gluteus medius, maximus and minimus, rectus abdominus, obliques, transversus abdominus
- Lower extremity
 - Tibialis anterior, peroneal muscles
- WEAK HYPOTONIC, INHIBITED or WEAK with repetitive stretch

Tonic/Postural VS Phasic in the Upper Extremity

TONIC/POSTURAL

- Slow Twitch
 - Oxidative
 - More capillaries-Red fibers
 - Phylogenetically older
 - Flexor or postural muscles
 - Typically one joint muscles
- Neck/Shoulder
 - Scalenes, subscapularis, flexors of upper extremity, levator scapulae, upper trapezius, sternocleidomastoid, pectorals, suboccipitals, cervical erector spinae, latissimus dorsi*

SHORT, TIGHT ,FACILITATED with repetitive • stress

PHASIC

- Fast Twitch
 - Glycolytic pathway
 - Low capillaries-White fibers
 - Phylogenetically younger
 - Generally extensor muscles
 - Typically two-joint muscles
- Neck/Shoulder
 - Infraspinatus, deltoid, deep cervical flexors, extensors of upper extremity, middle & lower trapezius, serratus anterior, rhomboids, supraspinatus, latissimus dorsi*,

WEAK HYPOTONIC, INHIBITED or WEAK with repetitive stretch

The Upper & Lower Crossed Syndrome

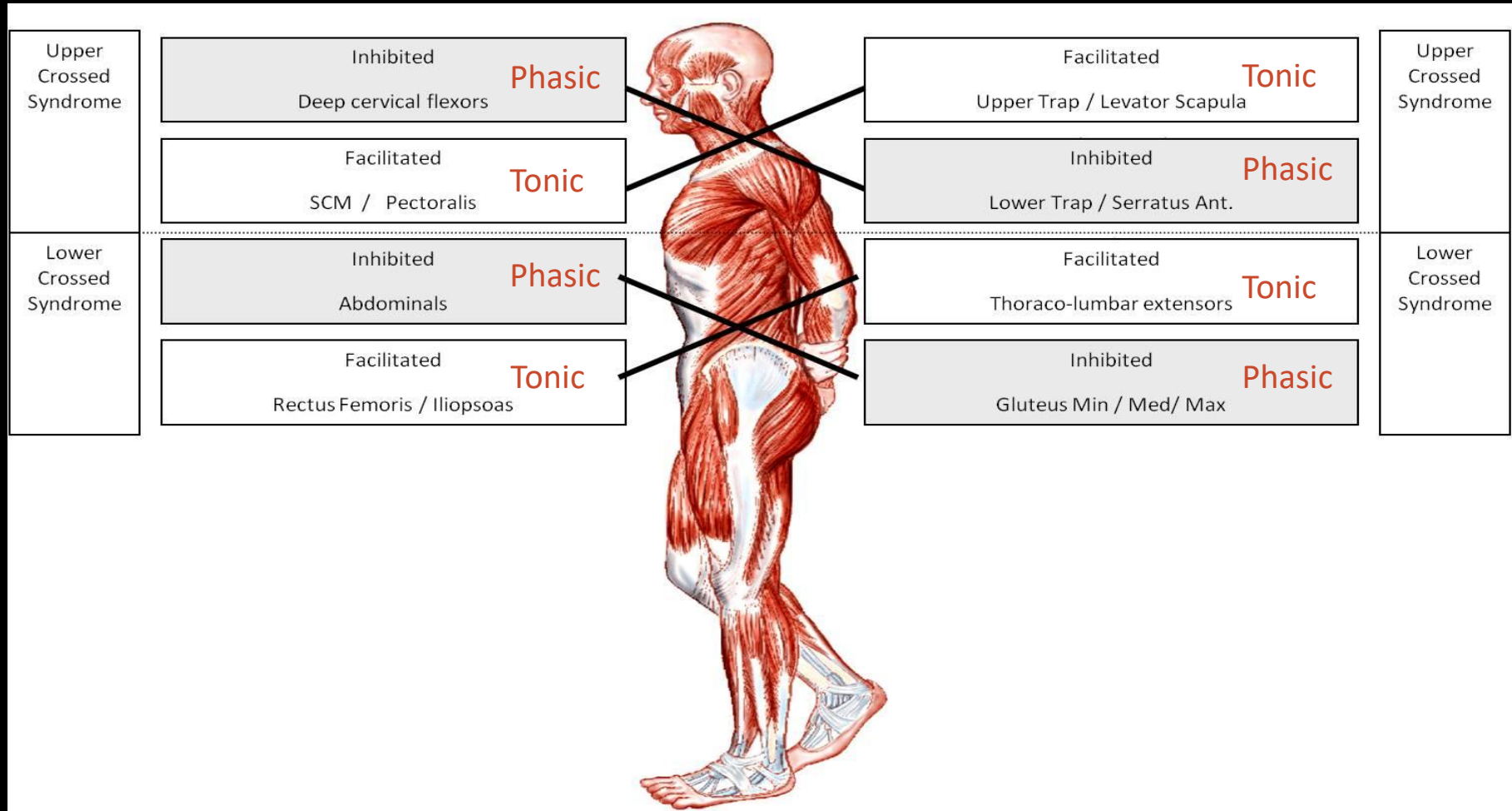
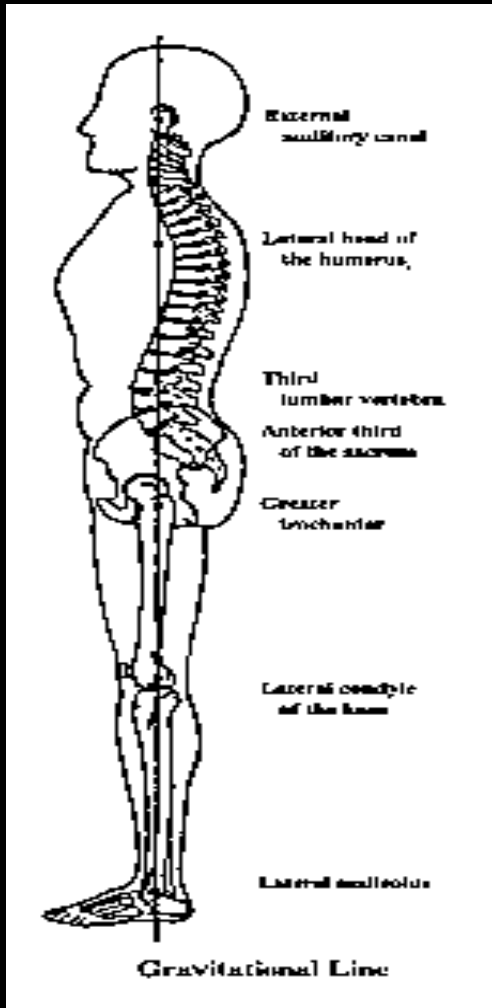


Figure 1 : Janda's Muscle Imbalance Syndromes

Begin by Observing Posture

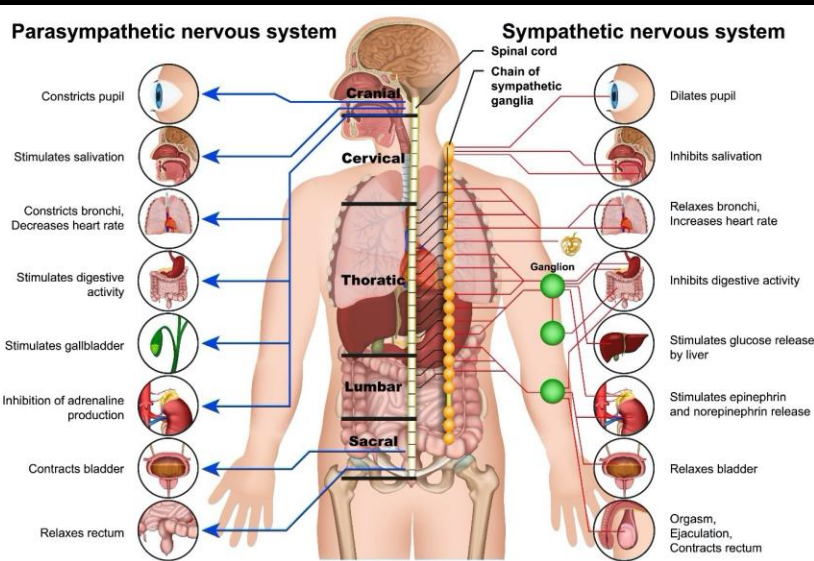
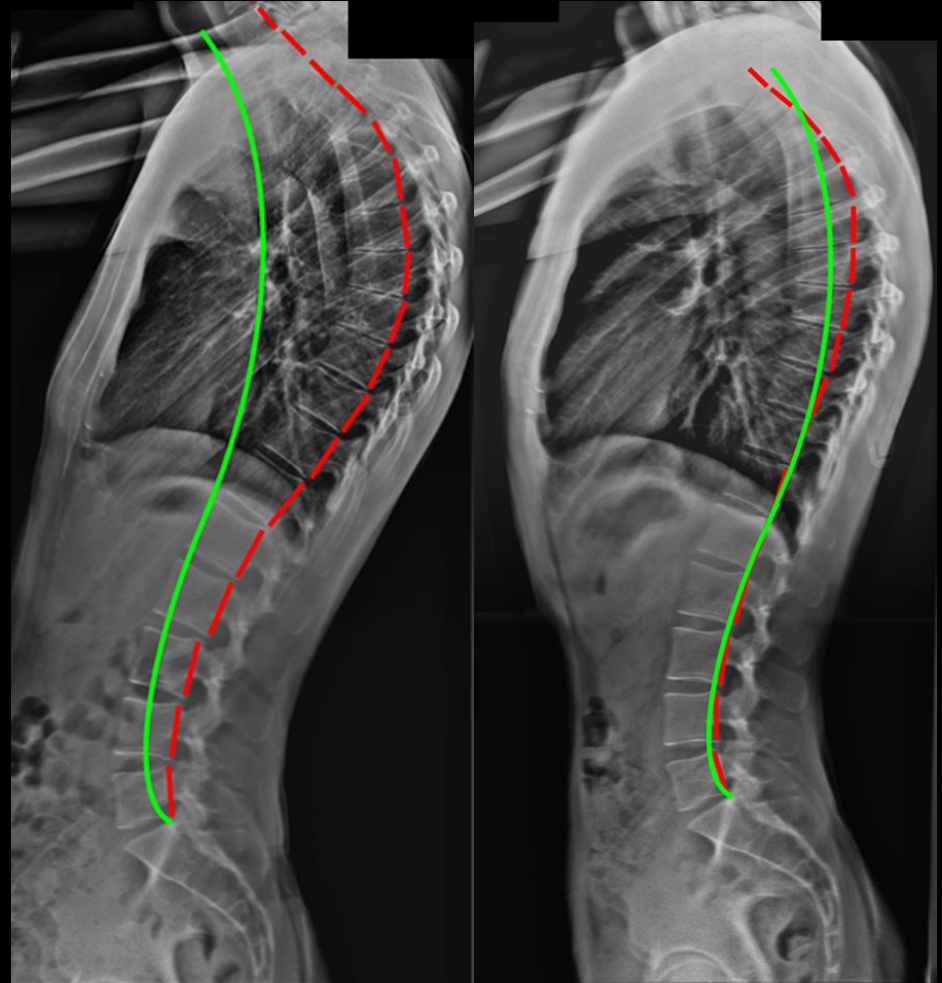


- **Gravitational Line**

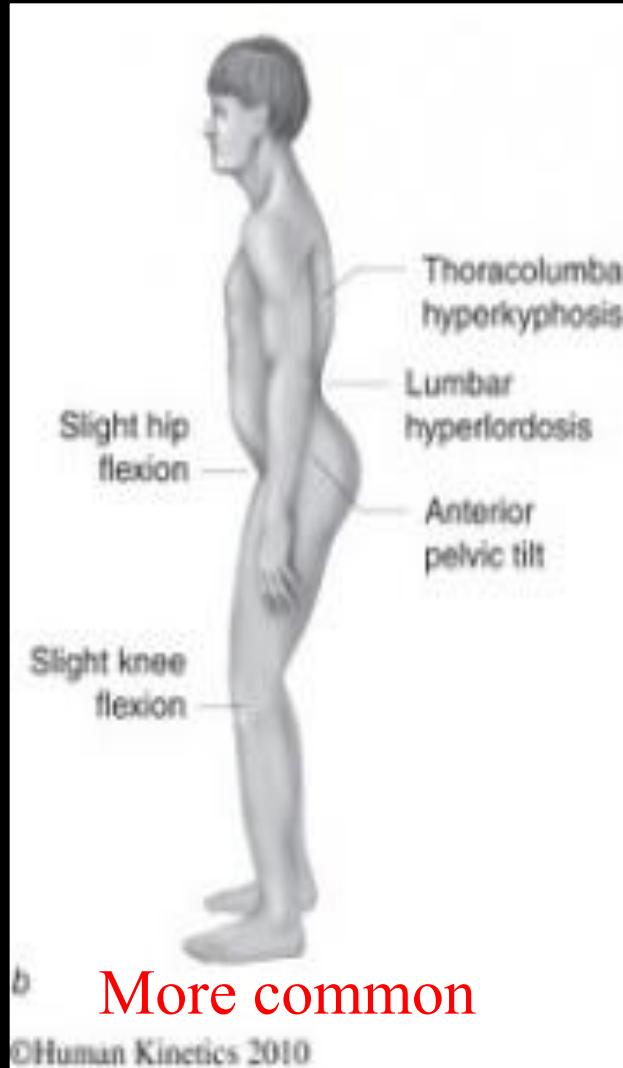
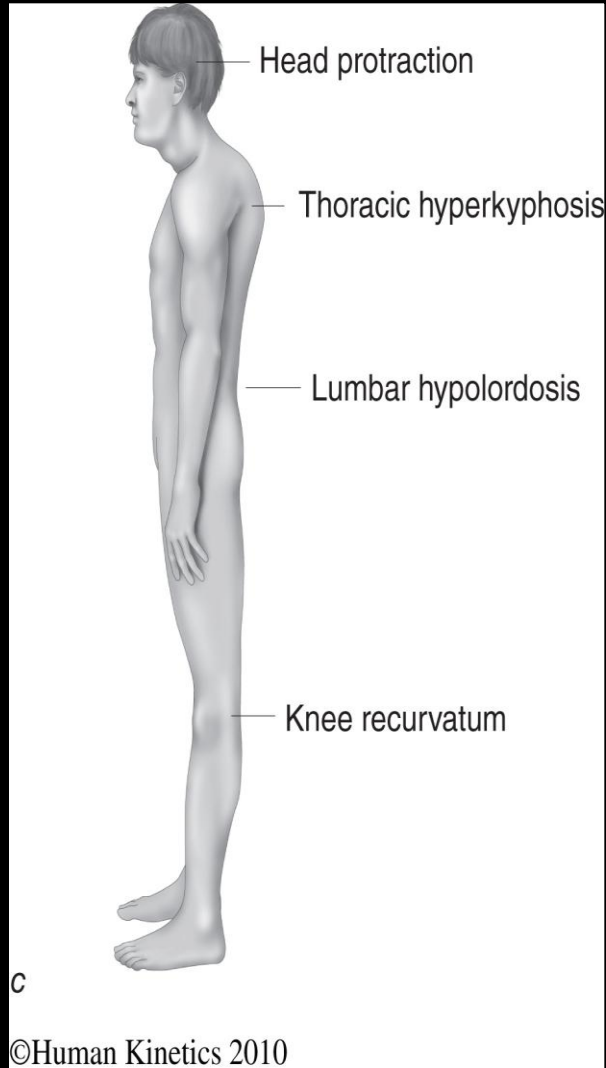
- External Auditory Meatus
- Lateral head of the humerus
- Body of L3
- Anterior third of the sacrum
- Greater Trochanter
- Lateral condyle of the knee
- Lateral malleolus

Posture and Viscera

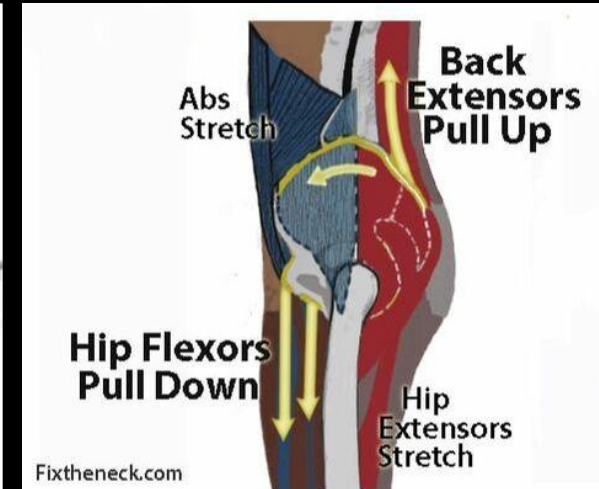
- Respiration
- Thoracic inlet
- Thoracic spine
 - facilitated spinal segments



LOWER CROSSED SYNDROME Posture



More common



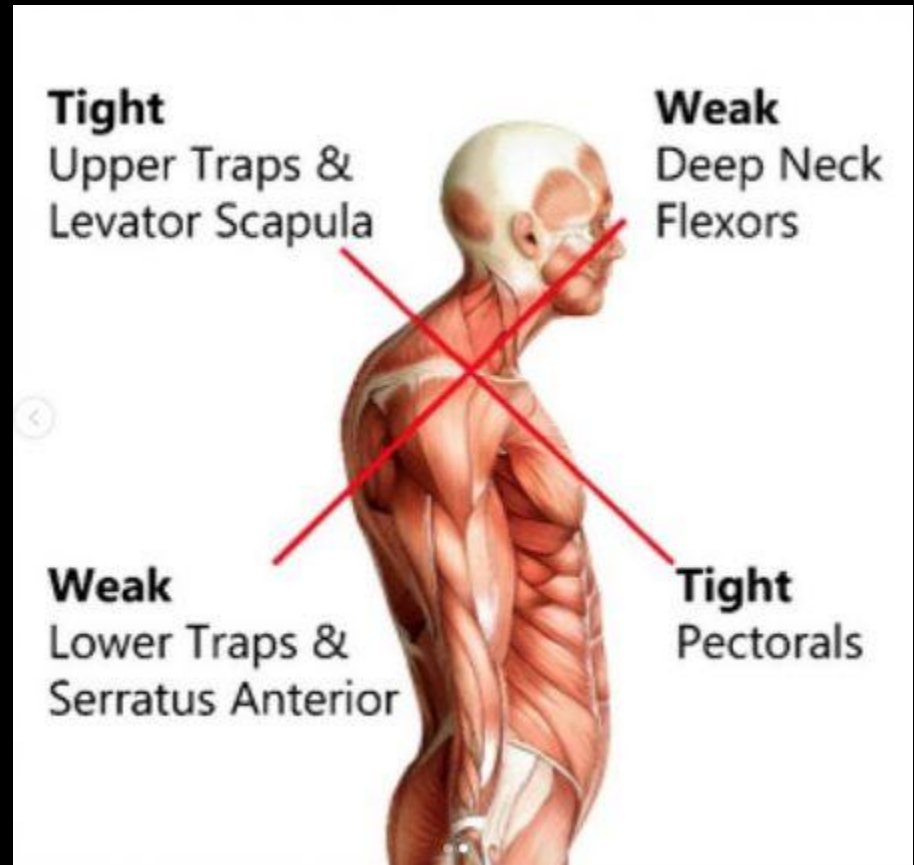
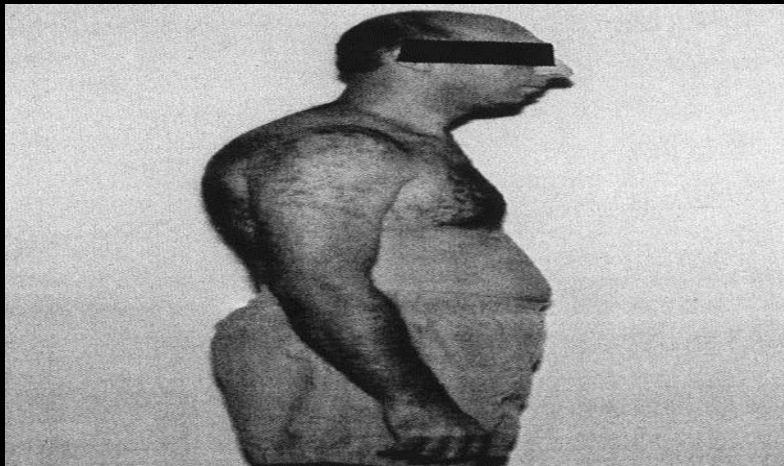
- spinal curves
- Leg lengths
- Tilt of the pelvis
- Gait
- Lower extremity muscle hypertrophy or atrophy

Assessment and Treatment of Muscle Imbalance- the Janda Approach (Page, Frank, Lardner)

Postural Changes that Affect Muscle Imbalance

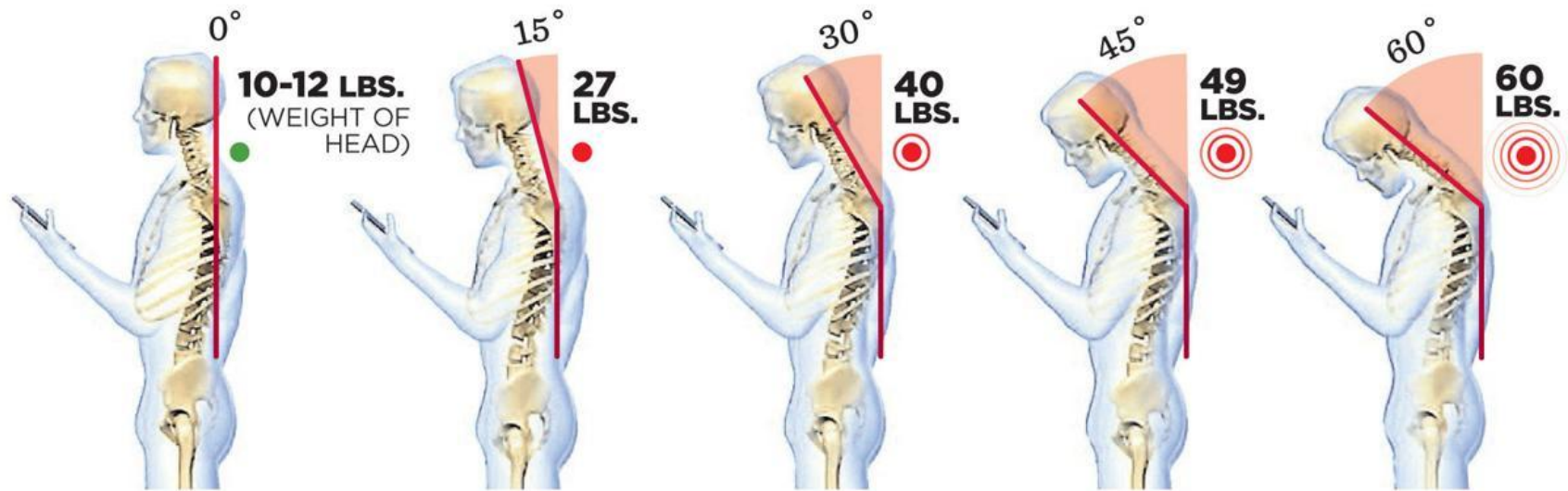
- **Upper Crossed**

- Forward Head Posture
- Extension of the OA and upper C-spine
- Increased kyphosis of the C/T junction (Hump at base of the neck)
- Protracted, elevated and internally rotated shoulders



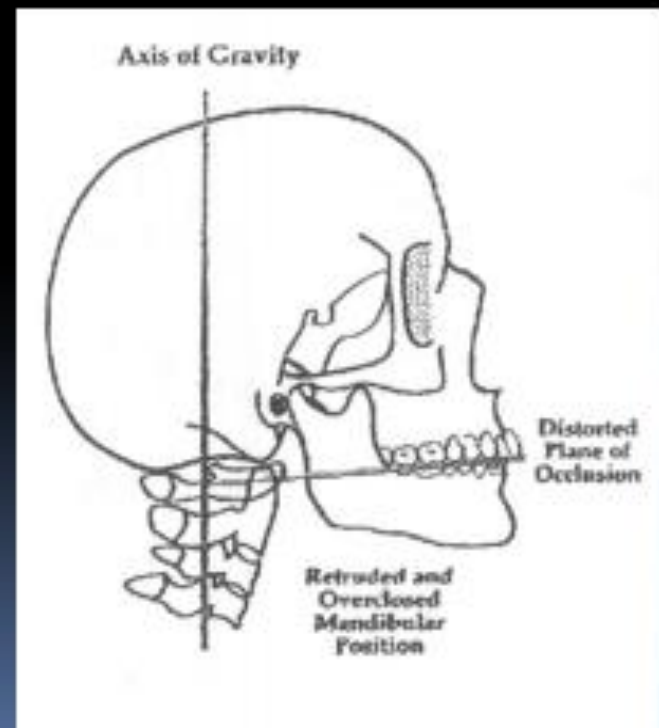
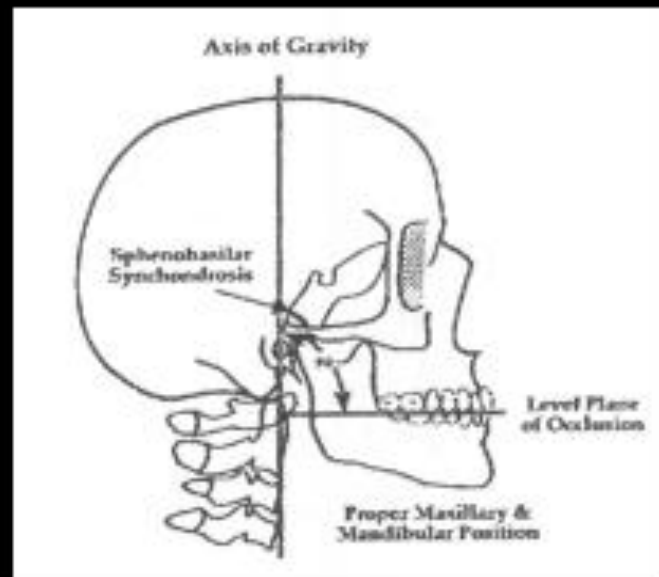
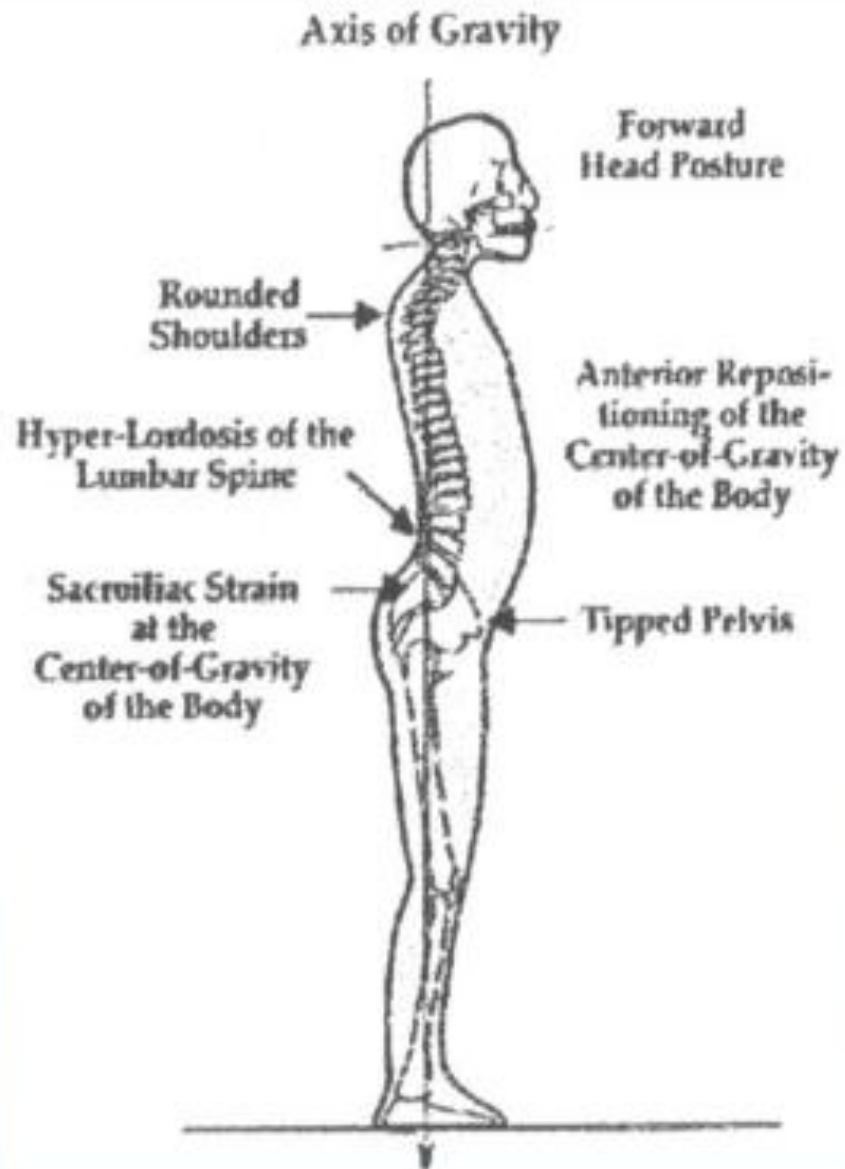
WATCH YOUR PHONE POSTURE

The pressure on the spine dramatically increases when the head is tilted forward. This stress to the spine can lead to early wear, tear and possible surgeries, according to research by Kenneth Hansraj in the National Library of Medicine.

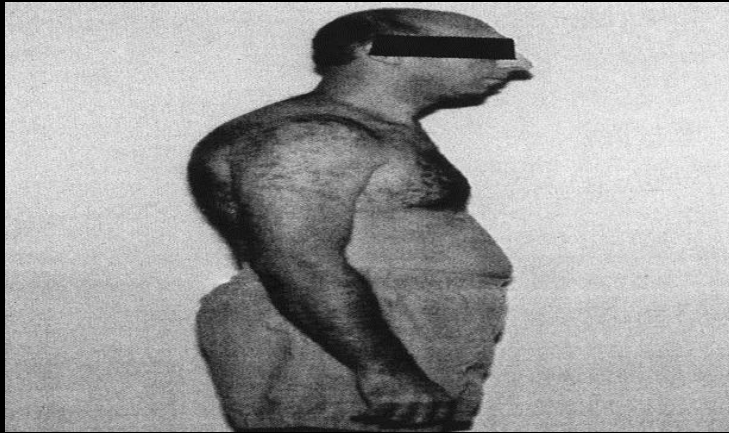


SOURCE: SURGICAL TECHNOLOGY INTERNATIONAL

People on average spend 2-4 hours a day looking at their cell phone. This equals 700-1,400 hours a year with poor spinal mechanics leading to loss of a normal cervical curve with the resulting stresses leading to wear, tear and degeneration.

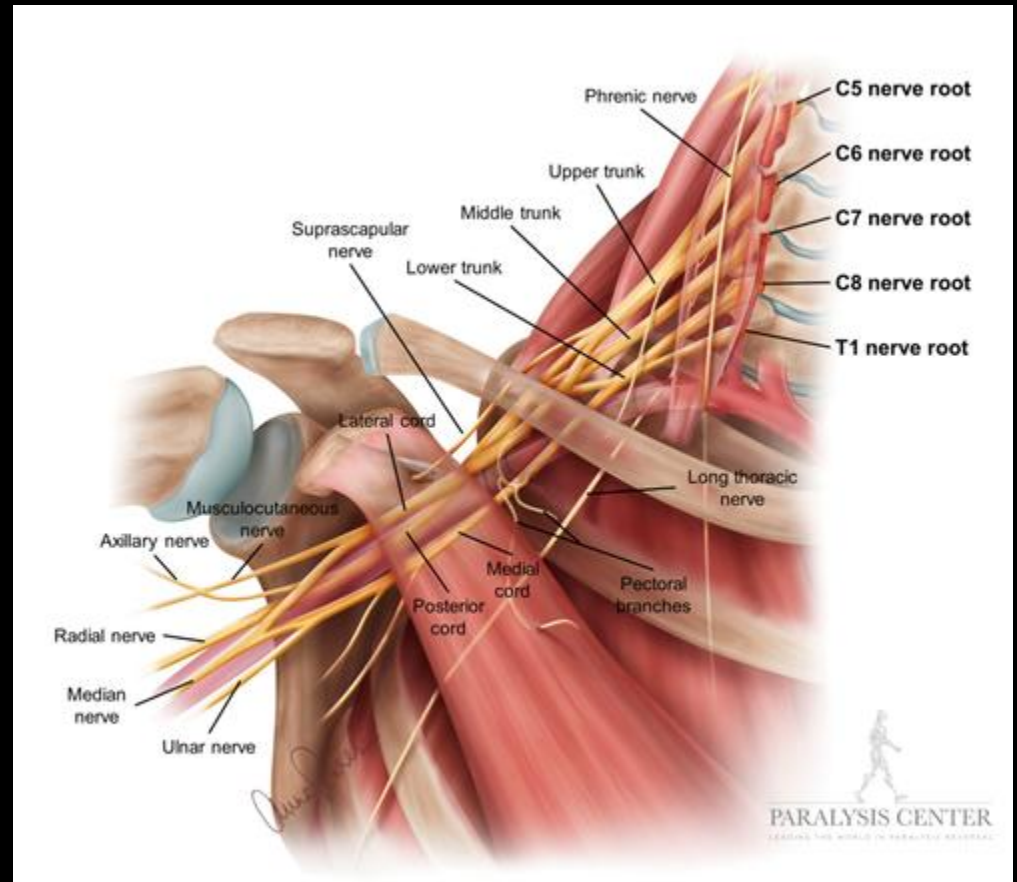


Bracial Plexus restriction



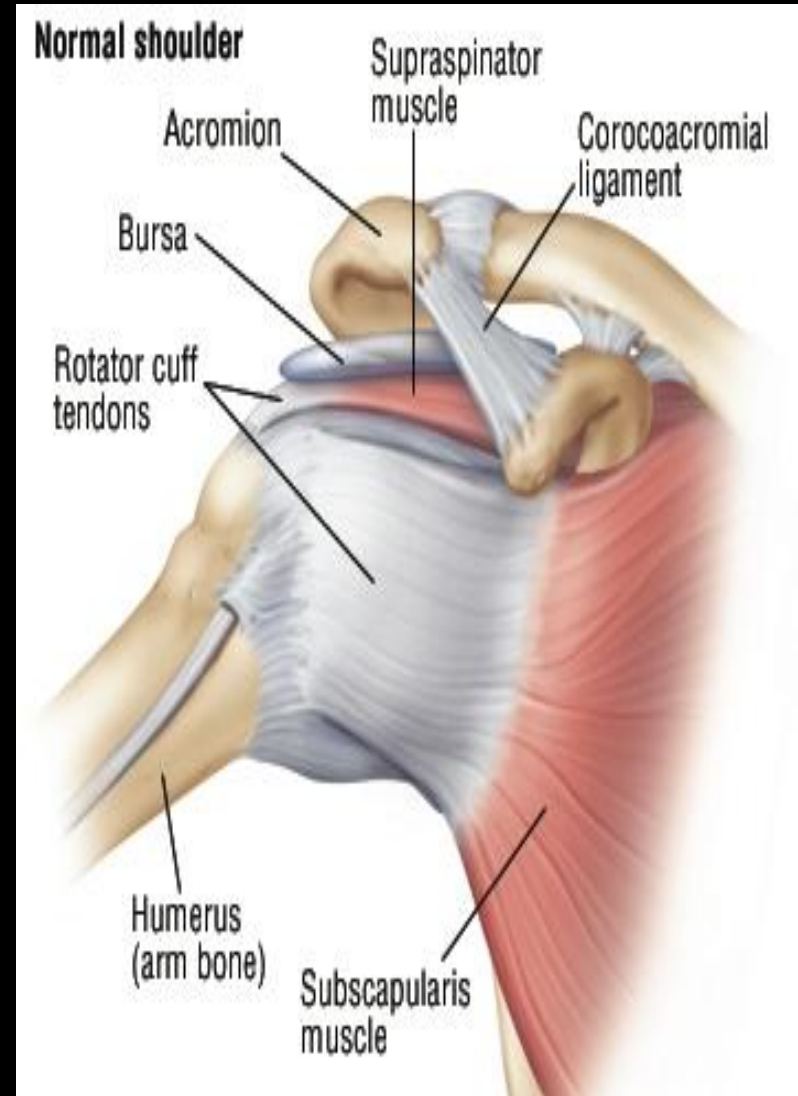
Nerve network that travels from the spine to the upper extremity

1. Anterior and Middle Scalene
Tonic
 - C2-7 - First rib
2. Clavical
3. Pectoralis minor **Tonic**



Subacromial Impingement

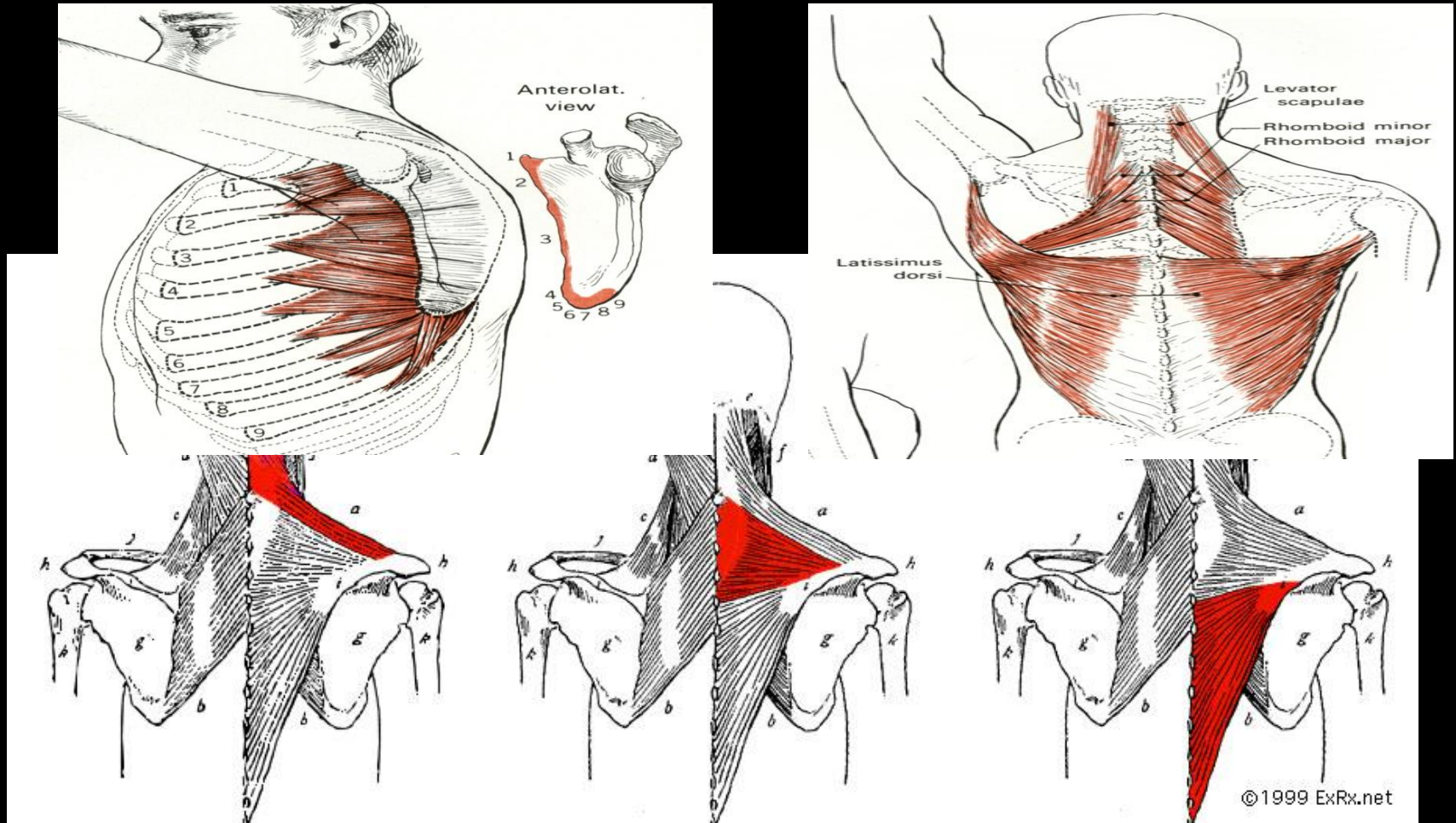
- Pts with impingement have less subacromial space (SAS) on MRI
 - Even though anatomically similar
 - Have more upward movement of the humerus
- Muscle imbalance affects shoulder movement and can decrease the SAS
 - Rotator cuff and deltoid
 - Rotator cuff is supposed to stabilize the humerus against the glenoid



Subacromial Impingement

- 44-65% of shoulder complaints
- **Structural impingement**
 - Physical loss of area in subacromial space
 - Bony, inflammation
- **Functional impingement**
 - Functional loss of subacromial space from “altered scapulohumeral mechanics resulting from glenohumeral instability and muscle imbalance”
 - Superior migration of the humeral head
- **Impingement occurs when the structures in the subacromial space become compressed**
 - Supraspinatus most at risk leading to degenerative tears

Shoulder Muscle Imbalance



Shoulder Impingement Muscle Imbalance

- Pectoralis minor hypertonicity
 - Limits scapular upward rotation, posterior tilt
 - Decreases SAS
- Patients with impingement EMG
 - Increased upper trapezius
 - Decrease lower trapezius, serratus anterior activity
- Lower trapezius
 - Important stabilizer
 - Decreases lower trapezius or increased upper trapezius activity
 - Leads to altered scapular rotation
 - Upward migration of the glenohumeral joint
 - Decreases SAS

Suspect Upper Crossed Syndrome in Patients with...

- Diagnoses of
 1. Upper neck, back or shoulder pain
 2. Shoulder Impingement syndrome
 3. Thoracic outlet syndrome
 4. Carpal Tunnel Syndrome
 5. TMJ dysfunction
 6. Chronic headaches
- Pain returns despite OMT
- Recurrent “knots” or trigger points

Review

- “when the manipulative intervention has achieved maximum mobility, the question remains, how is it maintained?”
 - “the obvious answer is an appropriate exercise program”
- Why?
 - “A disturbance of musculoskeletal function initiates a series of events beginning with stimulation to mechanoreceptors and nociceptors resulting in afferent neural activity initiating a variety of reflexes at the cord, brainstem and cortical levels. The final common pathway is the alpha motor neuron that stimulates the muscle fiber to contract and through the gamma system, the muscle spindle to adapt, resulting in the alteration in muscle tone”
 - “Interruption and reprogramming of this vicious cycle contributes to improvement of overall muscle tone and maintenance of balance”



Assessment and Treatment of the Upper & Lower Crossed Syndrome

The Janda ‘Recipe’

- Sensorimotor balance training
- Stretch before strengthen
- Re-educate movement pattern
 - Quality of movement emphasized.
- Restore Length, strength and control of muscle function



GOALS OF TREATMENT OF MUSCLE IMBALANCES

1. Goal of an exercise program is to
2. “maintain the enhanced functional capacity of the musculoskeletal system that has been achieved by appropriate manual medicine intervention”
3. Restoration of normal posture & movement patterns
4. Alleviation of pain & dysfunction
5. Protection of the osteoarticular system
6. Reduction of strain placed on joint capsules & ligaments
 - Appropriate muscle balance of agonists and antagonists provides shock absorption

STEP 1

PROPRIOCEPTIVE BALANCE

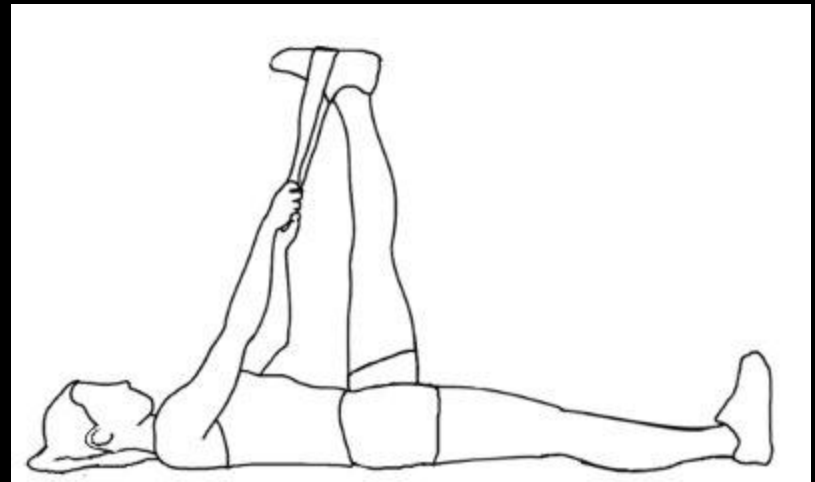
1. Proprioceptive balance is a key component to muscle firing pattern sequencing
2. Our orientation in space is dependent on three systems, the vestibular, visual, and sensory (stance our only contact with the ground)
3. Restoration of proprioceptive balance is the first stage of an exercise program.
4. Should look at not only if they can do it or not- but amount of pre-shift and excessive activity of knee, tibialis anterior or toes indicating poor proprioception
5. Proprioception may also be a pain gate



Step 2

Stretch Tonic/Facilitated Muscles

- Stretching to **alleviate** myofascial **hypertonicity** of tonic and facilitated muscles
- Stretching to **relieve** **pseudoparesis** of the phasic muscles



Step 3

Muscle re-education/ strengthening

- Teaching **correct muscle firing patterns** to weak muscles
- **Strengthen weak muscles**



Treat the CAUSE of the Muscle Imbalance

“Treatment of the dysfunctional postural alignment yields more lasting results than treatment of the secondary recurrent somatic dysfunction (myofascial trigger points, isolated muscle imbalances)”

FOM 3 p. 448

- Etiology and treatment involves:
 - Neuro-reflexive spinal cord component
 - Tight muscles need to be relaxed
 - Inhibited muscles need to be neurologically stimulated
 - A Viscoelastic component
 - Tight muscles must be manually lengthened
 - OMT should address both components- MET does!

The “Dirty Half Dozen”

- Research by Phillip Greenman DO in 1996
 - 183 patients with “failed lower back syndrome” despite ‘standard of care’
 - Found 6 osteopathic structural exam findings common to these patients
 - 178 (97.3%) of the patients had one or more of these
 - **Muscle imbalance was the most prevalent in 90%**
- 1. Non-neutral (type II) Lumbar dysfunction**
 - Usually lower lumbar, T/L junction; Mostly FRS
 - 2. Pubic symphysis dysfunction**
 - 3. Sacral base posterior**
 - Backward torsion or extended sacrum
 - 4. Innominate Shears**
 - 5. Leg Length discrepancy with un-level sacral base**
 - 6. Muscle imbalance**
 - Trunk and Lower extremities

Diagnostic tests and Exercise Prescription (self treatment) for Lower Crossed Syndrome

TREATMENT – Lower Crossed

TREATMENT- Address somatic dysfunction and Stretch Facilitated/Tonic Muscles

1. Diagnose & treat articular Somatic Dysfunction using the Mitchell Muscle Energy model.

This will address significant articular SD that may be causing a False (+) failure of a LCS Test, additionally, Muscle Energy to Sacrum, pelvis and Lumbar spine will also address many of these Postural/Tonic muscles via both neural and viscoelastic mechanisms.

2. Evaluate and treat with Muscle Energy the Postural-Tonic muscles

This should eliminate or decrease their inhibition of the phasic muscles.

3. Teach patient Home exercise program:

- a. Sensorimotor/Proprioception Balance Training
- b. Stretch lower body Facilitated/Tonic muscles using Muscle Energy
 - Iliopsoas
 - Rectus Femoris
 - Tensor Fascia Lata
 - Hip Adductors
 - Piriformis
 - Hamstrings
 - Quadratus Lumborum
 - Gastrocnemius/Soleus
- c. Re-education of movement pattern/strengthening of weak phasic muscles

Proprioceptive Balance Deficits

1. Instruct patient how to start at Level 1 (lowest level of difficulty)
– hold for up to 30 sec. switch feet, repeat
2. If 30 Sec. is achieved with each foot, move them up to next level of difficulty. If not, the test is done
3. Patient may rest briefly between levels to prevent false positive test due to fatigue



Difficulty levels:

Stage 1: One leg standing, arms down, eyes open

Stage 2: One leg standing, arms crossed, eyes open

Stage 3: One leg standing, arms down, eyes closed

Stage 4: One leg standing, arms crossed, eyes closed

Muscle Retraining/Strengthening

Patient practices the level of the
failed balance test

Pelvic Tilt & Rotation Test “Pelvic Clock”

- a. **Tilt Pelvis Posterior** (Decrease lordosis -Umbilicus toward 12 O’Clock/ tip bowl toward face)
- b. **Tilt Pelvis Anterior** (Increase lordosis -Umbilicus toward 6 O’Clock/ tip bowl toward feet)
- c. **Rotate Pelvis Right** (Move Umbilicus toward 9 O’Clock about 1-2 inches / tip bowl toward Right-ASIS stay level)
- d. **Rotate Pelvis Left** (Move Umbilicus toward 3 O’Clock about 1-2 inches / tip bowl toward Left-ASIS stay level)



Muscle Retraining/Strengthening

Patient practices the level of the
failed pelvic clock test

Pelvic Tilt with Hip Abduction Test

1. Patient Supine / knees bent, feet on table
2. Physician stands on eye dominant side / Thumbs on inferior aspect of ASIS's to monitor
3. Instruct patient to tilt pelvis posteriorly (toward 12 O'Clock/ tip bowl toward face)
4. Tell patient to hold the pelvis in that position while they let their knees GENTLY fall laterally. (The motion is passive thigh abduction, the adductors are undergoing eccentric contraction- the abductor muscles are not actively contracting to pull the thighs apart)
5. Note if one or both ASIS moves inferiorly (signifies patient unable to maintain posterior pelvic tilt)



Muscle Retraining/Strengthening

Patient practices the level of the
failed hip abduction test

Pelvic Tilt with Heel slide Hip Extension test

Procedure:

1. Patient Supine / Both knees bent, feet on table
2. Physician stands facing side of patient- monitor at both ASIS
3. Alternatively you can place your cephalad hand under the lumbar spine to monitor the lumbar lordosis
4. From neutral, patient tilts pelvis (toward 12 O'Clock) (which will flatten the lumbar lordosis)
5. Instruct patient to maintain pelvic tilt while sliding their heel along table to straighten knee and hip (hip extension)
6. Monitor for any dropping of the ASIS or increase in lumbar lordosis
7. Repeat with the other leg



Muscle Retraining/Strengthening

Retrain:

Retraining should only happen after stretching of the facilitated or hypertonic muscles.

1. Have patient lie supine and monitor themselves at the ASIS.
2. They contract their abs and lift up to 12'oclock position and slowly slide one leg down until the point when the ASIS drops.
3. Resume the start position and try again with the ultimate goal of hip extension with a stable ASIS.

Hip Extension Firing Pattern

The normal firing pattern for proper hip extension is

1. Hamstrings,
2. Gluteus Maximus
3. Contralateral Lumbar Erector Spinae,
4. Ipsilateral Lumbar Erector Spinae.
5. Significant findings include delayed or absent firing of the Gluteus Maximus and substitution by hamstrings and lumbar erector spinae.

Procedure:

1. Patient lies prone
2. Monitor at the hamstrings, glutes, and bilateral lumbar erector spinae
3. Instruct the patient to slowly lift their leg off the table (hip extension)
4. Observe for the firing pattern of the muscles



Iliopsoas/Anterior Hip Capsule

SELF STRETCH-

- find yourself in a low lunge- internally rotate the femur on the side to be stretched- try for a posterior pelvic tilt (12 O'clock)- contract your gluteal muscles on the same side- stretch both sides for symmetry



Rectus Femoris Stretch

SELF STRETCH-

- Flex the knee to the barrier, either against a table or using an ipsilateral hand to grasp the foot- tilt the pelvis in a posterior pelvic tilt (12 O'clock)- additional stretch by flexing the opposite knee- repeat both sides for symmetry



Hip Abduction Firing Pattern

The normal firing pattern for proper hip abduction is

1. Gluteus medius
2. Tensor fascia lata
3. Quadratus Lumborum
4. Lumbar erector spinae
5. Significant findings include late firing of the gluteus medius and early firing of the TFL resulting in internal rotation and flexion of the hip.

Procedure:

1. The patient is side-lying on the contralateral side you wish to test
2. Monitor at the TFL, Gluteus Medius, Quadratus Lumborum, and Lumbar Erector spinae
3. Instruct the patient to slowly abduct their leg off the table
4. Observe for the firing pattern of the muscles.



Muscle Retraining/Strengthening

Retraining of Gluteus Minimus/Gluteus Medius

1. Found to be weak or non-firing or firing out of pattern on the HIP ABDUCTION TEST with minimal improvement after stretching the antagonist musculature of the Adductors.
2. Patient lies in a later recumbent position with knees flexed 45-60 degrees
3. Patient should place one hand just below the iliac crest at the attachment of gluteus medius to feel for the contraction
4. Keeping the feet together, patient then lifts the top knee towards the ceiling and lowers after holding for several seconds
5. Perform 10-20 and rest- repeating 2-3 times



Muscle Retraining/Strengthening

Retraining of Gluteus Maximus

1. Have the patient lie in prone
2. Dorsiflex the ankle so the toes rest on the table of the side to be strengthened
3. Contract the Quads to lift the knee
4. Contract the gluteus maximus
5. Lift the foot off the table and plantarflex the ankle and hold for 10 seconds
6. Replace the leg back down in the same manner
7. Can advance to lifting the leg higher



Quadratus Lumborum Assessment

Assessment: standing (lateral sidebend) or sidelying sidebend. Look for differences in sidebending motion. The QL may be tight as a compensation for a weak gluteus medius

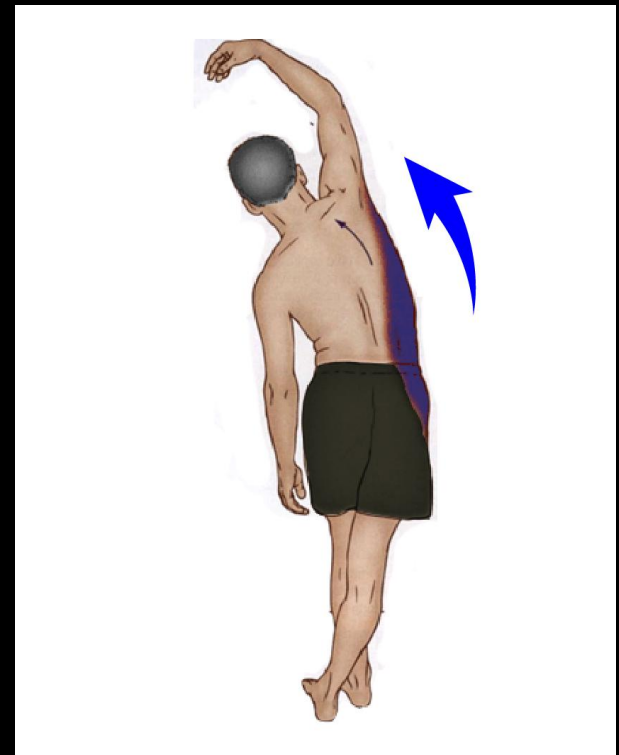
1. Pt is sidelying opposite the side that is being evaluated
2. Cephalad hand stabilizes the 12th rib
3. Caudad hand grasps the LE and abducts to relax QL, then allows the leg to drop towards the floor to assess length
4. Compare side to side
5. Utilize Muscle energy in this position by having the patient in this position push their leg up and after isometric contraction and relaxation take the leg into the next barrier towards the floor. The goal is to gain length between the 12th rib and the iliac crest (the attachment points of the QL).



Quadratus Lumborum Stretch

SELF STRECH-

1. Standing- raise arm on the same side you wish to stretch-
2. lean to the opposite side-
3. for more stretch cross you legs and push the hip you are stretching out to the side



Hamstring Assessment

- Assessment: as previously learned with Mitchell Model- usually very tight if Gluteus Maximus is weak
- Patient is lying supine- cephalad hand stabilizes the opposite ASIS- caudad hand grasps the distal tibia
- Flex the hip and monitor for hamstring hypertonicity- test each side
 - Adding more adduction tests the lateral hamstrings and abducting tests more of the medial hamstrings



Hamstring

SELF STRETCH

1. the Patient lies supine and flexes the hip- they can clasp behind the thigh with either their hands or a towel looped around their foot- they can perform their own MET by pulling their leg towards the table (engaging the hamstrings) and moving into the next hip flexion barrier.
2. This can also be in done standing and bending forward
3. This can also be done seated and hinging forward from the hips- keeping the spine straight- contracting the quads



Adductor Assessment

Assessment:

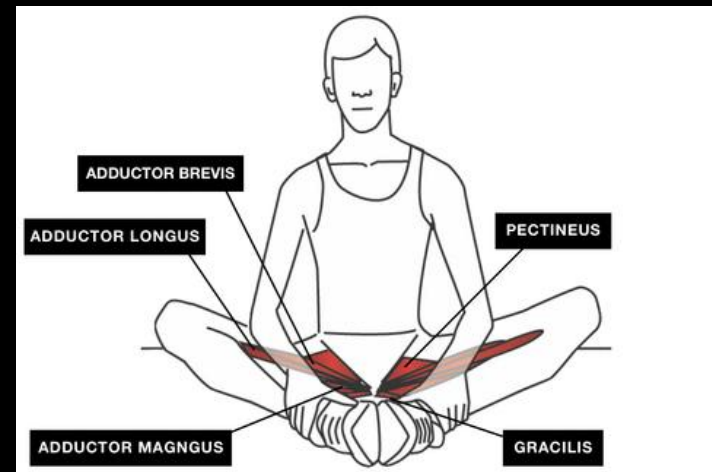
1. Patient lies supine
2. For the LONG adductors (crosses two joints)- standing in between the patients legs, use one hand to stabilize the opposite leg and the other hand to grasp the ankle and abduct the leg into the barrier.
3. For the SHORT adductors (cross one joint)- have the patient flex the knee and rest the foot on the table. Cephalad hand will stabilize the ASIS- Caudad hand will rest on the medial knee and slowly abduct the hip to the barrier.
4. You should be able to get to 45 degrees of abduction without pelvic movement
5. FOR BOTH utilize MET by having the patient contract the ADDUCTORS and take up the slack, increasing the ABDUCTION of the LE.



Hip Adductors

SELF STRETCH-

1. In a seated position- bring the plantar surface of your feet together as your hips ABDUCT.
2. Alternatively, you can sit in a wide leg seat (or standing) and fold forward- hinging from the hips and keeping the spine straight



Gastronemius/Soleus Assessment

Assessment:

1. patient lies supine
2. Distract the calcaneus caudally until no further slack can be taken up- hold this position with one hand cupping the heel while the other hand wraps around the distal edge of the foot and induces dorsiflexion. Compare side to side.
 - Tightness in these muscles can create abnormal foot mechanics and shift the bodies center of gravity anteriorly causing compensatory engagement of the erector spinae muscles, perpetuating low back pain.
3. You may perform muscle energy in this position, taking up the slack in dorsiflexion

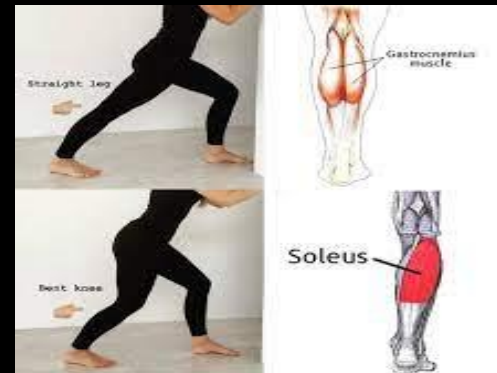
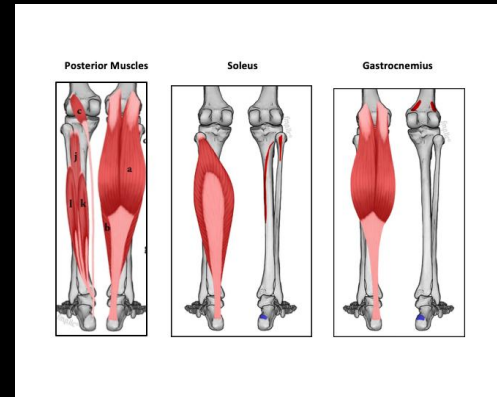


Gastronemius/Soleus

SELF STRETCH

Gastrocnemius

- Have your patient perform self-stretches by standing facing a wall- arms resting on the wall.
- One foot is against the wall dorsiflexed with the other is about 1 foot back.
- Flex first the knee of the foot against the wall and hold for a stretch for 30 seconds.
- Then straighten that leg and flex the knee of the posterior foot.



Let's look back at our Patient now

- What else would you want to look for now?



- **VSS** Normal, normal gait with over-pronation and right flattened arch
- **Abdominal Exam** Normal
- **Neuro:** LE strength +5/5 in all Myotomes, Sensation intact to light touch in all Dermatomes, UE/LE DTR +2/4 no clonus, SLR negative
- **MSK:** normal LS ROM, no step off, increased lordosis, anterior pelvic tilt hypertonic lumbar paraspinals, no CVA tenderness, no scoliosis
- **Osteopathic:** Right innominate anterior, R on L sacral torsion, bilateral hamstring and adductor hypertonicity worse on the right, bilateral iliopsoas hypertonicity worse on the right, L5ERSI, L1-4NSIRr, right tibial torsion, right fibular head posterior, right cuboid dropped.

Additional Findings

- Loss of proprioception
 - Gets to level 2 for 20 sec L
 - Gets to level 1 for 20 sec R
- Bilateral hypertonic hamstrings and adductors, ITB, rectus femoris, iliopsoas, lumbar erector spinae
- Weak abdominals, weak glutes (hip extension, hip abduction)
 - Tested by 6 tests we will do in LAB



Plan for our Patient

- **Treat Somatic Dysfunction** (which may be facilitating muscle imbalance)
- **Proprioception Retraining** (same as with lower crossed)
- **Stretch hypertonic/facilitated** postural/ Tonic muscles
- **Re-assess** 5 postural tests and see if there is improvement
 - If improved then phasic muscles are just inhibited
 - If not improved then they are truly weak
- **Muscle re-education/strengthening** Phasic muscles

Diagnostic tests and Exercise Prescription (self treatment) for Upper Crossed Syndrome

Treatment – Upper Crossed

1. Diagnose & treat Somatic Dysfunction in the OA, AA, C2-7, thoracic spine and associated ribs, using any techniques, as tolerated

- This should clear up any significant SD that may be causing a False (+) failure of a UCS Test

2. Evaluate the Postural-Tonic muscles bilaterally for tightness and stretch: This should eliminate or decrease their inhibition of the phasic muscles.

3. Teach patient Home exercise program:

- a. Sensorimotor/Proprioception Balance Training
- b. Stretch upper body Facilitated/Tonic muscles using Muscle Energy
 - Levator Scapula
 - Upper Trapezius and Sternocleidomastoid
 - Pectoralis major and minor
 - Latissimus dorsi
 - Posterior shoulder capsule
- c. Re-education of movement pattern/strengthening of weak phasic muscles
 - Cervical flexion retraining
 - Rhomboid and Lower Trapezius retraining

Cervical Flexion Test

Procedure:

1. Patient Supine and the physician sits or stands at side of table near patient's head
2. Ask patient to SLOWLY try to look at their feet
3. Chin should tilt toward chest and remain there as patient continues to raise head (Forehead should be more anterior than chin throughout the range of motion)



Failed test signifies:

- i. Inhibited/ weak: Deep cervical flexors **Phasic** (longus colli, longus capitus, rectus capitus anterior)
- ii. Due to:
 1. Dominance of SCM and scalenes (weakness of deep flexors leads to increased activity of these synergists)
 2. Hypertonicity of Cervical Erector Spinae **Tonic**

Muscle Retraining/Strengthening

1. **Cervical Flexion Retraining**
 - a. Have the patient seated with a hand palpating the superficial anterior cervical muscles
 - b. The patient should nod their head forward utilizing the deep neck flexors (without contracting superficial muscles)
 - c. Hold for 5-7 seconds completing 5-7 repetitions



Bilateral Shoulder Abduction Test

Procedure:

1. Patient Seated, with good upright posture, physician standing behind patient, hands monitoring the scapula
2. Make sure that the patient's abduction is not limited due to seated position. The starting position must be by their side. It is best to use a chair rather than a treatment table.
1. Begin by looking at the scapula- is one more protracted than another
2. Instruct patient to raise their arms alongside their head
3. Previous labs emphasized using this test to lateralize to the most dysfunctional side by noting which scapula started moving first. Both sides may be dysfunctional so describe what you feel on BOTH sides
4. Note which side protracts more
5. Have the patient do it again and note if there is early scapular elevation (before 60 degrees of shoulder abduction)
6. If the patient sidebends the thoracic spine and elevates the shoulder to achieve the motion- this is an indicator of significant dysfunction



LS/SCM/Upper Trapezius

SELF STRETCH

- a) In a seated position, flex, sidebend away and rotate towards (for SCM) the desired side to stretch. Stabilize the side to be stretched by grasping the chair to stabilize the scapular end.
- b) You can also stretch scalenes and upper trapz in this way- modified by altering the rotation (rotate away)
- c) The patient should sidebend away and rotate and chin tuck until the stretch is felt
- d) You can modify by looking inferiorly, to the middle and superiorly to stretch different fibers

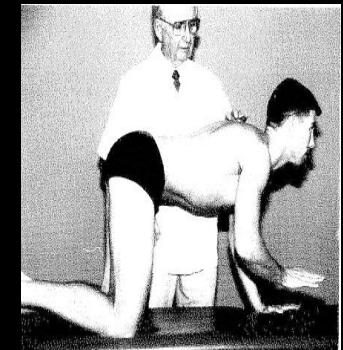
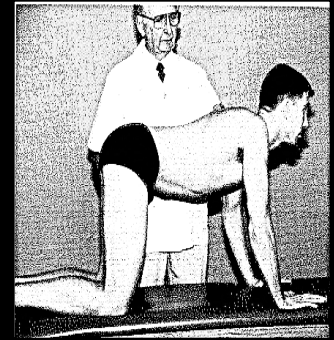


Scapular Stabilization

Procedure:

Patient on table – on hands & knees with elbows flexed enough to keep spine parallel (avoid spinal sagging or arching) to floor and in neutral position (see above) or against wall.

- Notice any winging or scapular elevation
 - Physician stands next to patient and monitors the inferomedial border of scapula being tested. (The arm that remains on the table)
 - First look and notice if there is already a difference between scapula
 - Instruct patient to support themselves with one arm while lifting the other off the table or wall a few inches-keeping palm parallel to the table or wall (see above pictures)
 - Physician observes and palpates for any winging of scapula on support side arm
5. Test other side



PECTORALIS MAJOR AND MINOR

SELF STRETCH

- a) The patient stands facing the wall and places both hands on the wall at shoulder height
- b) Slowly the patient turns away from the side to be stretched until a stretch is felt in the pectoral region
- c) You can also recommend the doorway stretch



Muscle Retraining/Strengthening

Rhomboid and Lower Trapezius

Retraining

- a) Have the patient seated with their feet on the floor
- b) Bring their hands and arms in front of them interlacing the fingers in front of the sternum
- c) The patient should attempt to pull their elbows backward (as though pulling the fingers apart) for 5-7 seconds and repeat 5-7 times to strengthen the Rhomboids.
- d) The patient may then bring their interlaced fingers overhead and attempt to pull elbows down toward the floor for 5-7 seconds for 5-7 times to strengthen the lower trapezius



Scapular Depression

Procedure:

1. Have the patient lie prone with the arm to be test flexed.
2. Monitor at the inferior medial border of the scapula

Look for contraction of the lower trapezius and instruct the patient to pull the scapula down and back against resistance.

Compare sides



SELF STRETCH- for LS/SCM/Upper Trapezius

- i. In a seated position, flex, sidebend away and rotate towards (for SCM) the desired side to stretch. Stabilize the side to be stretched by grasping the chair to stabilize the scapular end.
- ii. You can also stretch scalenes and upper trapz in this way- modified by altering the rotation (rotate away)
- iii. The patient should sidebend away and rotate and chin tuck until the stretch is felt
- iv. You can modify by looking inferiorly, to the middle and superiorly to stretch different fibers



Supine Bilateral Shoulder Flexion

Procedure:

1. Have the patient lie supine
2. Bring both of the patients arms overhead into shoulder flexion.
3. Assess the tension and degree of ability of shoulder flexion, and the ability to maintain a neutral lumbar spine (avoiding lumbar extension). Note any asymmetry.



LATISSIMUS DORSI

1. ***SELF STRETCH***

- a) Have the patient kneel, and bring the hands together and elbows together on a stool (also known as the prayer stretch)
- b) Patient stretches by dropping the chest to the floor and sitting back towards the heels
- c) Can also be done standing- but maintaining a flat back



Know for Lab and for the TEST!

Function of (should know origin/insertion too but will not be tested on it):

- Iliopsoas- hip flexion
- Rectus Femoris- hip flexion
- Tensor Fascia Lata- hip flexion, IR, ABD
- Lumbar Erector Spinae- spine extension, minor hip extension
- Quadratus Lumborum- ipsilateral sidebending, spine extension, minor hip extension
- Abdominals- Spine flexion
- Glut Max- hip extension
- Glut Medius and Minimus- hip ABD/IR
- Hip Adductors- hip Adduction
- Piriformis- Hip ER, above 90 degrees abduction
- Quadriceps (3 vastus)- knee extension
- Rectus femoris- Knee extension, hip flexion
- Gastroc/Soleus- Ankle plantar flexors
- Tibialis Anterior- Ankle dorsiflexor

Know for Lab and for the TEST!

- **Function of (should know origin/insertion too but will not be tested on it):**
 - Upper Trapezius- elevates the scapula
 - Lower/Middle Trapezius-retracts and rotates the glenoid superiorly and stabilize the scapula
 - Levator Scapulae- elevates the scapula
 - Scalenes- head sidebending and some rotation
 - Sternocleidomastoid- flexes the c-spine and sidebending and rotation
 - Pectoralis Major-shoulder adduction
 - Pectoralis Minor- shoulder adduction and some flexion
 - Deltoid- abduction, flexion and extension of the shoulder
 - Rhomboids- scapular retraction
 - Serratus Anterior- scapular protraction
 - Subscapularis- shoulder internal rotation
 - Infraspinatus- shoulder external rotation
 - Supraspinatus- shoulder abduction to 15 degrees
 - Deep Neck Flexors (Longus Colli, Longus Capitus)- cervical flexion
 - Latissimus Dorsi-extends, adducts humerus, rotates scapula inferiorly

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